Namespace LlamaLogic.Packages

Classes

DataBasePackedFile

Represents a Maxis DBPF format package file (🔒)

ResourceFileTypeAttribute

Specifies the file type of a package resource type independent of storage within a DataBasePackedFile

ResourceFileTypeFormatAttribute

Specifies the extension and mime type of a file type of a package resource independent of storage within a DataBasePackedFile

ResourceToolingMetadataAttribute

Specifies that the ResourceType is tooling metadata and not intended for use by the game

SmartSimUtilities

Provides utility methods specific to The Sims 4

Structs

ResourceKey

Represents the unique key of a resource within a DataBasePackedFile

Enums

CompressionMode

Represents the compression mode to use when setting the content of a resource in a DataBasePackedFile

ResourceFileType

The file type of a package resource independent of storage within a DataBasePackedFile

ResourceKeyOrder

Specifies the order in which a ResourceKey sequence should be processed

ResourceType

A type of resource within a DataBasePackedFile

Enum CompressionMode

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents the compression mode to use when setting the content of a resource in a DataBasePackedFile

```
public enum CompressionMode
```

Fields

`Auto = 0`

The library will choose whether and how to compress the data or not

`CallerSuppliedInternal = 6`

⚠️ The data will not be processed for compression by the library and the resource will be marked as having been compressed using Maxis' internal method (use with caution)

`CallerSuppliedStreamable = 7`

⚠️ The data will not be processed for compression by the library and the resource will be marked as having been compressed using Maxis' streamable method (use with caution)

`CallerSuppliedZLib = 8`

⚠️ The data will not be processed for compression by the library and the resource will be marked as having been compressed using ZLIB (use with caution)

`ForceInternal = 4`

The data will be compressed using Maxis' internal method

`ForceOff = 1`

The data will not be compressed

`[Obsolete("This setting can no longer be used and will be removed in a future version. If you need to force a compression, use a different force member of this enum.", true)] ForceOn = 2`

Deprecated: do not use

`ForceZLib = 3`

The data will be compressed using ZLib

`SetDeletedFlag = 5`

⚠️ The data will not be processed for compression by the library—thus callers will need to invoke ZLibCompress(ReadOnlyMemory<byte>) or ZLibCompressAsync(ReadOnlyMemory<byte>, CancellationToken) themselves and pass the result to Set(ResourceKey, ReadOnlyMemory<byte>, CompressionMode) or SetAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, CancellationToken), respectively, if they intend for the content to be compressed with ZLib despite being flagged—and the deleted flag will be set (use with caution)

Remarks

Important: While this library does support the older compression methods, it does not have a complete catalog of how to do so properly for every resource type automatically (Auto). If you want to be sure that you're compressing a resource correctly, utilize GetExplicitCompressionMode(ResourceKey) with a package the game is currently accepting.

Warning: The Sims 4 does not always expect or tolerate ZLib compression for certain types of resources, such as for world package files which ship with the game (see thepancake1's comment about this on the Creator Musings Discord for more information).

Class DataBasePackedFile

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents a Maxis DBPF format package file (🔒)

```
public sealed class DataBasePackedFile : IAsyncDisposable, IDisposable
```

Inheritance

object ← DataBasePackedFile

Implements

IAsyncDisposable, IDisposable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Streams and Disposal

When a package is opened by this class from a Stream, the class assumes ownership of the Stream. Manipulating the Stream afterward may result in undefined behavior. The class will dispose of the Stream when it is, itself, disposed.

To use the Save(bool, ResourceKeyOrder) or SaveAsync(bool, ResourceKeyOrder, CancellationToken) methods, the package must have been opened from a Stream which is writeable. This can also be determined by the caller in advance of attempting to use them by checking the value of the CanSaveInPlace property. However, for ease of use in web frameworks like ASP.NET Core, the CopyTo(Stream, ResourceKeyOrder) and CopyToAsync(Stream, ResourceKeyOrder, CancellationToken) methods do *not* require that the Stream which they are passed is seekable.

Lazy Loading

When a package is first opened using this class, the package index is read but the content of the package resources is not. All of the retrieval methods of this class have async counterparts for callers who prefer not to be blocked by I/O operations which may be required when retrieving resource content.

Because resource names (when applicable) are determined by their content, this class will not index resource names until they are first referenced, either explicitly by calling GetNames() or GetNamesAsync(CancellationToken), or implicitly by calling any of the other methods which deal with resource names. Afterward, the names are cached for the lifetime of the package instance and updated as the caller sets the content of resources or deletes them.

Thread Safety

All properties and methods of this class, including async methods, are kept thread-safe with internally managed locks. Async methods which accept cancellation tokens will honor signals so long as they have not progressed to mutating the state of the package.

Compression

Decompression of resource content is handled automatically by this class.

When retrieving resource content, this class will automatically decompress it using either proprietary Maxis internal or streaming compression, or ZLib compression. When saving resources, this class will always use ZLib compression. For more details, see the remarks for the CompressionMode enumeration.

Resource content is compressed in memory when it is set to minimize the amount of memory used while operating on a package.

Deleted Entries

This class honors the deleted entry flag in the package index under normal circumstances. If a caller attempts to retrieve the content of a resource flagged as deleted, the class will throw a FileNotFoundException by default. All retrieval methods have an optional `force` bool parameter which, when true, will cause the class to attempt to retrieve the content of a resource flagged as deleted (this is not guaranteed to work).

Constructors

DataBasePackedFile()

Initializes a new, blank DataBasePackedFile

```
public DataBasePackedFile()
```

DataBasePackedFile(Stream)

Initializes a DataBasePackedFile from the specified `stream` (🗑️ 💤)

```
public DataBasePackedFile(Stream stream)
```

Parameters

`stream` Stream

DataBasePackedFile(string, bool)

Initializes a DataBasePackedFile from the specified `path` (🗑️ 💤)

```
public DataBasePackedFile(string path, bool forReadOnly = true)
```

Parameters

`path` string

`forReadOnly` bool

Properties

CanSaveInPlace

Gets whether the package can be saved in place

```
public bool CanSaveInPlace { get; }
```

Property Value

bool

Count

Gets the number of resources

```
public int Count { get; }
```

Property Value

int

CreationTime

Gets/sets when the package was created

```
public DateTimeOffset CreationTime { get; set; }
```

Property Value

DateTimeOffset

FileVersion

Gets/sets the version of the format of the file

```
public Version FileVersion { get; set; }
```

Property Value

Version

Remarks

This class is intended to work with versions of the Maxis DBPF format up to version `2.1` and going back as far as possible. While it will permit the caller to specify any version they like, it does not guard against potential negative consequences which may result from such packages actually being used. The default version for newly created packages is currently `2.1`.

this[ResourceKey]

Gets/sets the content of a resource with the specified `key`

```
public object? this[ResourceKey key] { get; set; }
```

Parameters

`key` ResourceKey

Property Value

object

Keys

Gets a list of keys for all the resources in the package

```
public IReadOnlyList<ResourceKey> Keys { get; }
```

Property Value

IReadOnlyList<ResourceKey>

Unused1

Gets/sets the four bytes in the package header starting at index 14h

```
public ReadOnlyMemory<byte> Unused1 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused2

Gets/sets the four bytes in the package header starting at index 20h

```
public ReadOnlyMemory<byte> Unused2 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused3

Gets/sets the twelve bytes in the package header starting at index 30h

```
public ReadOnlyMemory<byte> Unused3 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused4

Gets/sets the four bytes in the package header starting at index 3Ch

```
public ReadOnlyMemory<byte> Unused4 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

Unused5

Gets/sets the twenty-four bytes in the package header starting at index 48h

```
public ReadOnlyMemory<byte> Unused5 { get; set; }
```

Property Value

ReadOnlyMemory<byte>

UpdatedTime

Gets/sets when the package was last updated

```
public DateTimeOffset UpdatedTime { get; set; }
```

Property Value

DateTimeOffset

UserVersion

Gets/sets the version of the user (presumably the user agent, actually? -- is Maxis versioning human beings?)

```
public Version UserVersion { get; set; }
```

Property Value

Version

Methods

ContainsKey(ResourceKey)

Gets whether the package contains a resource with the specified `key`

```
public bool ContainsKey(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

bool

ContainsKeyAsync(ResourceKey, CancellationToken)

Gets whether the package contains a resource with the specified `key` asynchronously

```
public Task<bool> ContainsKeyAsync(ResourceKey key, CancellationToken cancellationToken
= default)
```

Parameters

`key` ResourceKey

`cancellationToken` CancellationToken

Returns

Task<bool>

CopyTo(Stream, ResourceKeyOrder)

Copies the package in binary format to the specified `destination` (🔄 🏃)

```
public void CopyTo(Stream destination, ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`destination` Stream

`resourceKeyOrder` ResourceKeyOrder

Exceptions

ArgumentNullException

`destination` is null

ArgumentException

`destination` is not writeable

CopyToAsync(Stream, ResourceKeyOrder, CancellationToken)

Copies the package in binary format to the specified `destination` asynchronously (🔄 🏃)

```
public Task CopyToAsync(Stream destination, ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`destination` Stream

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task

Exceptions

ArgumentNullException

`destination` is null

ArgumentException

`destination` is not writeable

Delete(ResourceKey)

Deletes the resource with the specified `key` and returns true if it was found; otherwise, false

```
public bool Delete(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

bool

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

DisposeAsync()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources asynchronously.

```
public ValueTask DisposeAsync()
```

Returns

ValueTask

A task that represents the asynchronous dispose operation.

~DataBasePackedFile()

Called by the finalizer

```
protected ~DataBasePackedFile()
```

ForEach(ResourceKeyOrder, Predicate<ResourceKey>, Action<ResourceKey, ReadOnlyMemory<byte>>, CancellationToken)

Processes undeleted resources in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAction`

```
public void ForEach(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Action<ResourceKey, ReadOnlyMemory<byte>> processResourceAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAction` Action<ResourceKey, ReadOnlyMemory<byte>>

`cancellationToken` CancellationToken

ForEachAsync(ResourceKeyOrder, Predicate<ResourceKey>, Func<ResourceKey, ReadOnlyMemory<byte>, Task>, CancellationToken)

Processes undeleted resources in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAsyncAction`

```
public Task ForEachAsync(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Func<ResourceKey, ReadOnlyMemory<byte>, Task> processResourceAsyncAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAsyncAction` Func<ResourceKey, ReadOnlyMemory<byte>, Task>

`cancellationToken` CancellationToken

Returns

Task

ForEachRaw(ResourceKeyOrder, Predicate<ResourceKey>, Action<ResourceKey, ReadOnlyMemory<byte>>, CancellationToken)

Processes undeleted raw resources (meaning that the library will not decompress a resource for you if it is compressed) in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAction`

```
public void ForEachRaw(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Action<ResourceKey, ReadOnlyMemory<byte>> processResourceAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAction` Action<ResourceKey, ReadOnlyMemory<byte>>

`cancellationToken` CancellationToken

ForEachRawAsync(ResourceKeyOrder, Predicate<ResourceKey>, Func<ResourceKey, ReadOnlyMemory<byte>, Task>, CancellationToken)

Processes undeleted raw resources (meaning that the library will not decompress a resource for you if it is compressed) in the specified `keyOrder` if they satisfy the specified `keyPredicate` using the specified `processResourceAsyncAction`

```
public Task ForEachRawAsync(ResourceKeyOrder keyOrder, Predicate<ResourceKey> keyPredicate,
Func<ResourceKey, ReadOnlyMemory<byte>, Task> processResourceAsyncAction, CancellationToken
cancellationToken = default)
```

Parameters

`keyOrder` ResourceKeyOrder

`keyPredicate` Predicate<ResourceKey>

`processResourceAsyncAction` Func<ResourceKey, ReadOnlyMemory<byte>, Task>

`cancellationToken` CancellationToken

Returns

Task

FromPathAsync(string, bool)

Initializes a DataBasePackedFile asynchronously from the specified `path` (🗑️ 💤)

```
public static Task<DataBasePackedFile> FromPathAsync(string path, bool forReadOnly = true)
```

Parameters

`path` string

`forReadOnly` bool

Returns

Task <DataBasePackedFile>

FromStreamAsync(Stream, CancellationToken)

Initializes a DataBasePackedFile asynchronously from the specified `stream` (🗑️ 💤)

```
public static Task<DataBasePackedFile> FromStreamAsync(Stream stream, CancellationToken
cancellationToken = default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task <DataBasePackedFile>

Get(ResourceKey, bool)

Gets the content of a resource with the specified `key`

```
public ReadOnlyMemory<byte> Get(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

ReadOnlyMemory<byte>

GetAllSizes(Predicate<ResourceKey>?)

Gets the size of the content of each resource in the package indexed by key

```
public IReadOnlyDictionary<ResourceKey, int> GetAllSizes(Predicate<ResourceKey>? predicate
= null)
```

Parameters

`predicate` Predicate<ResourceKey>

An optional predicate to filter the keys

Returns

IReadOnlyDictionary<ResourceKey, int>

GetAllSizesAsync(Predicate<ResourceKey>?, CancellationToken)

Gets the size of the content of each resource in the package indexed by key, asynchronously

```
public Task<IReadOnlyDictionary<ResourceKey, int>> GetAllSizesAsync(Predicate<ResourceKey>?
predicate = null, CancellationToken cancellationToken = default)
```

Parameters

`predicate` Predicate<ResourceKey>

An optional predicate to filter the keys

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<IReadOnlyDictionary<ResourceKey, int>>

GetAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` asynchronously

```
public Task<ReadOnlyMemory<byte>> GetAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

GetCount()

Gets the number of resources

```
public int GetCount()
```

Returns

int

GetCountAsync(CancellationToken)

Gets the number of resources asychronously

```
public Task<int> GetCountAsync(CancellationToken cancellationToken = default)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task<int>

GetData(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a DataModel

```
public DataModel GetData(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

DataModel

GetDataAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a DataModel asynchronously

```
public Task<DataModel> GetDataAsync(ResourceKey key, bool force = false, CancellationToken
cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<DataModel>

GetDdsAsPng(ResourceKey, bool)

Gets the content of a resource that is a DirectDraw Surface with the specified `key`, encoded as a PNG

```
public ReadOnlyMemory<byte> GetDdsAsPng(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

ReadOnlyMemory<byte>

GetDdsAsPngAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource that is a DirectDraw Surface with the specified `key`, encoded as a PNG, asynchronously

```
public Task<ReadOnlyMemory<byte>> GetDdsAsPngAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

GetExplicitCompressionMode(ResourceKey)

Gets the CompressionMode value to pass to Set methods to ensure they will compress the content of the resource specified by `key` in the same way it is currently compressed in the package

```
public CompressionMode GetExplicitCompressionMode(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

CompressionMode

GetKeys(ResourceKeyOrder)

Gets a list of keys for all the resources in the package

```
public IReadOnlyList<ResourceKey> GetKeys(ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`resourceKeyOrder` ResourceKeyOrder

Returns

IReadOnlyList<ResourceKey>

GetKeysAsync(ResourceKeyOrder, CancellationToken)

Gets a list of keys for all the resources in the package asynchronously

```
public Task<IReadOnlyList<ResourceKey>> GetKeysAsync(ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task<IReadOnlyList<ResourceKey>>

GetKeysByName(string)

Gets a list of keys for resources with the specified `name`

```
public IReadOnlyList<ResourceKey> GetKeysByName(string name)
```

Parameters

`name` string

Returns

IReadOnlyList<ResourceKey>

GetKeysByNameAsync(string, CancellationToken)

Gets a list of keys for resources with the specified `name` asynchronously

```
public Task<IReadOnlyList<ResourceKey>> GetKeysByNameAsync(string name, CancellationToken
cancellationToken = default)
```

Parameters

`name` string

`cancellationToken` CancellationToken

Returns

Task<IReadOnlyList<ResourceKey>>

GetModFileManifest(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a ModFileManifestModel

```
public ModFileManifestModel GetModFileManifest(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

ModFileManifestModel

GetModFileManifestAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a ModFileManifestModel asynchronously

```
public Task<ModFileManifestModel> GetModFileManifestAsync(ResourceKey key, bool force = false, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<ModFileManifestModel>

GetModelAsync<TModel>(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a `TModel` asynchronously

```
public Task<TModel> GetModelAsync<TModel>(ResourceKey key, bool force = false, CancellationToken cancellationToken = default) where TModel : IModel<TModel>
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task <TModel>

Type Parameters

`TModel`

A model used to browse and modify certain types of resources

Exceptions

ArgumentException

`TModel` cannot deal with resources of the type specified by `key`

GetModel<TModel>(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a `TModel`

```
public TModel GetModel<TModel>(ResourceKey key, bool force = false) where TModel
: IModel<TModel>
```

Parameters

`key` ResourceKey

`force` bool

Returns

TModel

Type Parameters

`TModel`

A model used to browse and modify certain types of resources

Exceptions

ArgumentException

`TModel` cannot deal with resources of the type specified by `key`

GetNameByKey(ResourceKey)

Gets the name for the resource with the specified `key`

```
public string? GetNameByKey(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

string

GetNameByKeyAsync(ResourceKey, CancellationToken)

Gets the name for the resource with the specified `key` asynchronously

```
public Task<string?> GetNameByKeyAsync(ResourceKey key, CancellationToken cancellationToken
= default)
```

Parameters

`key` ResourceKey

`cancellationToken` CancellationToken

Returns

Task <string>

GetNames()

Gets the names for all the resources in the package

```
public IReadOnlyList<string> GetNames()
```

Returns

IReadOnlyList<string>

GetNamesAsync(CancellationToken)

Gets the names for all the resources in the package asynchronously

```
public Task<IReadOnlyList<string>> GetNamesAsync(CancellationToken cancellationToken
= default)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task<IReadOnlyList<string>>

GetRaw(ResourceKey, bool)

Gets the raw content of a resource with the specified `key`, meaning that the library will not decompress it for you if it is compressed

```
public ReadOnlyMemory<byte> GetRaw(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

ReadOnlyMemory<byte>

GetRawAsync(ResourceKey, bool, CancellationToken)

Gets the raw content of a resource with the specified `key` asynchronously, meaning that the library will not decompress it for you if it is compressed

```
public Task<ReadOnlyMemory<byte>> GetRawAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

GetSize(ResourceKey)

Gets the size of the content of the resource with the specified `key`

```
public int GetSize(ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

int

GetSizeAsync(ResourceKey, CancellationToken)

Gets the size of the content of the resource with the specified `key` asynchronously

```
public Task<int> GetSizeAsync(ResourceKey key, CancellationToken cancellationToken
= default)
```

Parameters

`key` ResourceKey

`cancellationToken` CancellationToken

Returns

Task<int>

GetStringTable(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a StringTableModel

```
public StringTableModel GetStringTable(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

StringTableModel

GetStringTableAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a StringTableModel asynchronously

```
public Task<StringTableModel> GetStringTableAsync(ResourceKey key, bool force = false,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<StringTableModel>

GetText(ResourceKey, bool)

Gets the content of a resource with the specified `key` as a string

```
public string GetText(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

string

GetTextAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as a string asynchronously

```
public Task<string> GetTextAsync(ResourceKey key, bool force = false, CancellationToken
cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task <string>

GetTranslucentJpegAsPng(ResourceKey, bool)

Gets the content of a resource that is a TS4 proprietary translucent JPEG based with the specified `key`, reintegrated into RGBA and encoded as a PNG

```
public ReadOnlyMemory<byte> GetTranslucentJpegAsPng(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

`force` bool

Returns

ReadOnlyMemory<byte>

GetTranslucentJpegAsPngAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource that is a TS4 proprietary translucent JPEG based with the specified `key`, reintegrated into RGBA and encoded as a PNG, asynchronously

```
public Task<ReadOnlyMemory<byte>> GetTranslucentJpegAsPngAsync(ResourceKey key, bool force = false, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

`force` bool

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

GetXml(ResourceKey, bool)

Gets the content of a resource with the specified `key` as an XDocument

```
public XDocument GetXml(ResourceKey key, bool force = false)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

Returns

XDocument

GetXmlAsync(ResourceKey, bool, CancellationToken)

Gets the content of a resource with the specified `key` as an XDocument asynchronously

```
public Task<XDocument> GetXmlAsync(ResourceKey key, bool force = false, CancellationToken
cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`force` bool

true to get the content of the resource even if it has been marked as deleted; otheriwse, false (default)

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<XDocument>

InternalCompress(ReadOnlyMemory<byte>, out bool)

Compresses the specified `memory` in the manner of Maxis' internal algorithm, returning the result

```
[SuppressMessage("Design", "CA1021: Avoid out parameters")]
public static ReadOnlyMemory<byte> InternalCompress(ReadOnlyMemory<byte> memory, out
bool wasCompressed)
```

Parameters

`memory` ReadOnlyMemory<byte>

`wasCompressed` bool

Returns

ReadOnlyMemory<byte>

InternalDecompress(ReadOnlyMemory<byte>)

Decompresses the specified `memory` in the manner of Maxis' internal algorithm, returning the result

```
public static ReadOnlyMemory<byte> InternalDecompress(ReadOnlyMemory<byte> memory)
```

Parameters

`memory` ReadOnlyMemory<byte>

Returns

ReadOnlyMemory<byte>

LoadAll(bool, CompressionMode)

Loads all resources in the package into memory

```
public void LoadAll(bool force = false, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`force` bool

true to get the content of the resources even if they has been marked as deleted; otheriwse, false (default)

`compressionMode` CompressionMode

This parameter is now ignored

LoadAllAsync(bool, CompressionMode, CancellationToken)

Loads all resources in the package into memory

```
public Task LoadAllAsync(bool force = false, CompressionMode compressionMode =
CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`force` bool

true to get the content of the resources even if they has been marked as deleted; otheriwse, false (default)

`compressionMode` CompressionMode

This parameter is now ignored

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

LoadNames()

Loads the names of all the resources in the package (a no-op if the names are already loaded)

```
public void LoadNames()
```

LoadNamesAsync(CancellationToken)

Loads the names of all the resources in the package asynchronously (a no-op if the names are already loaded)

```
public Task LoadNamesAsync(CancellationToken cancellationToken = default)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task

Save(bool, ResourceKeyOrder)

Saves the package to the stream from which it was loaded

```
public void Save(bool unloadFromMemory = false, ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`unloadFromMemory` bool

`resourceKeyOrder` ResourceKeyOrder

Exceptions

InvalidOperationException

The package was not loaded from stream or the stream is not writeable

SaveAs(string, ResourceKeyOrder)

Saves the package to the specified `path` (🔄 🏃)

```
public void SaveAs(string path, ResourceKeyOrder resourceKeyOrder
= ResourceKeyOrder.Preserve)
```

Parameters

`path` string

`resourceKeyOrder` ResourceKeyOrder

SaveAsAsync(string, ResourceKeyOrder, CancellationToken)

Saves the package to the specified `path` asynchronously (🔄 🏃)

```
public Task SaveAsAsync(string path, ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`path` string

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task

SaveAsync(bool, ResourceKeyOrder, CancellationToken)

Saves the package to the stream from which it was loaded asynchronously

```
public Task SaveAsync(bool unloadFromMemory = false, ResourceKeyOrder resourceKeyOrder =
ResourceKeyOrder.Preserve, CancellationToken cancellationToken = default)
```

Parameters

`unloadFromMemory` bool

`resourceKeyOrder` ResourceKeyOrder

`cancellationToken` CancellationToken

Returns

Task

Exceptions

InvalidOperationException

The package was not loaded from stream or the stream is not writeable

Set(ResourceKey, IModel, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, IModel content, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` IModel

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Set(ResourceKey, ReadOnlyMemory<byte>, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, ReadOnlyMemory<byte> content, CompressionMode
compressionMode = CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` ReadOnlyMemory<byte>

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Set(ResourceKey, string, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, string content, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` string

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

Set(ResourceKey, XDocument, CompressionMode)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool Set(ResourceKey key, XDocument content, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`content` XDocument

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

SetAsync(ResourceKey, IModel, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, IModel content, CompressionMode compressionMode
= CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` IModel

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

SetAsync(ResourceKey, ReadOnlyMemory<byte>, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, ReadOnlyMemory<byte> content, CompressionMode
compressionMode = CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` ReadOnlyMemory<byte>

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

SetAsync(ResourceKey, string, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, string content, CompressionMode compressionMode
= CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` string

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

SetAsync(ResourceKey, XDocument, CompressionMode, CancellationToken)

Sets the `content` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetAsync(ResourceKey key, XDocument content, CompressionMode
compressionMode = CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`content` XDocument

The content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

SetPngAsDds(ResourceKey, ReadOnlyMemory<byte>)

Sets the content of a resource specified by `key` to a shuffled DirectDraw Surface based on `pngData`

```
public void SetPngAsDds(ResourceKey key, ReadOnlyMemory<byte> pngData)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

SetPngAsDdsAsync(ResourceKey, ReadOnlyMemory<byte>, CancellationToken)

Sets the content of a resource specified by `key` to a shuffled DirectDraw Surface based on `pngData` asycnhronously

```
public Task SetPngAsDdsAsync(ResourceKey key, ReadOnlyMemory<byte> pngData,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

SetPngAsTranslucentJpeg(ResourceKey, ReadOnlyMemory<byte>)

Sets the content of a resource specified by `key` to a TS4 proprietary translucent JPEG based on `pngData`

```
public void SetPngAsTranslucentJpeg(ResourceKey key, ReadOnlyMemory<byte> pngData)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

SetPngAsTranslucentJpegAsync(ResourceKey, ReadOnlyMemory<byte>, CancellationToken)

Sets the content of a resource specified by `key` to a TS4 proprietary translucent JPEG based on `pngData` asycnhronously

```
public Task SetPngAsTranslucentJpegAsync(ResourceKey key, ReadOnlyMemory<byte> pngData,
CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`pngData` ReadOnlyMemory<byte>

The encoded PNG data

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task

SetXml(ResourceKey, string, CompressionMode)

Sets the `xmlContent` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public bool SetXml(ResourceKey key, string xmlContent, CompressionMode compressionMode
= CompressionMode.Auto)
```

Parameters

`key` ResourceKey

The key of the resource

`xmlContent` string

The XML content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

Returns

bool

SetXmlAsync(ResourceKey, string, CompressionMode, CancellationToken)

Sets the `xmlContent` of a resource with the specified `key`, returning true if the resource was compressed; otherwise, false

```
public Task<bool> SetXmlAsync(ResourceKey key, string xmlContent, CompressionMode
compressionMode = CompressionMode.Auto, CancellationToken cancellationToken = default)
```

Parameters

`key` ResourceKey

The key of the resource

`xmlContent` string

The XML content of the resource

`compressionMode` CompressionMode

The compression mode to use for the resource

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<bool>

ZLibCompress(ReadOnlyMemory<byte>)

Compresses the specified `memory` using ZLib, returning the result

```
public static ReadOnlyMemory<byte> ZLibCompress(ReadOnlyMemory<byte> memory)
```

Parameters

`memory` ReadOnlyMemory<byte>

Returns

ReadOnlyMemory<byte>

ZLibCompressAsync(ReadOnlyMemory<byte>, CancellationToken)

Compresses the specified `memory` using ZLib asynchronously, returning the result

```
public static Task<ReadOnlyMemory<byte>> ZLibCompressAsync(ReadOnlyMemory<byte> memory,
CancellationToken cancellationToken = default)
```

Parameters

`memory` ReadOnlyMemory<byte>

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

ZLibDecompress(ReadOnlyMemory<byte>, int)

Decompresses the specified `memory` using ZLib, returning the result at the specified `sizeDecompressed`

```
public static ReadOnlyMemory<byte> ZLibDecompress(ReadOnlyMemory<byte> memory,
int sizeDecompressed)
```

Parameters

`memory` ReadOnlyMemory<byte>

`sizeDecompressed` int

Returns

ReadOnlyMemory<byte>

ZLibDecompressAsync(ReadOnlyMemory<byte>, int, CancellationToken)

Decompresses the specified `memory` using ZLib asynchronously, returning the result at the specified `sizeDecompressed`

```
public static Task<ReadOnlyMemory<byte>> ZLibDecompressAsync(ReadOnlyMemory<byte> memory,
int sizeDecompressed, CancellationToken cancellationToken = default)
```

Parameters

`memory` ReadOnlyMemory<byte>

`sizeDecompressed` int

`cancellationToken` CancellationToken

Returns

Task<ReadOnlyMemory<byte>>

Enum ResourceFileType

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

The file type of a package resource independent of storage within a DataBasePackedFile

```
public enum ResourceFileType
```

Fields

`[ResourceFileTypeFormat("bin", "application/octet-stream")] Binary = 0`

Binary

`[ResourceFileTypeFormat("dds", "image/vnd.ms-dds")] DirectDrawSurface = 1`

DDS Image

`[ResourceFileTypeFormat("png", "image/png")] DirectDrawSurfaceAsPortableNetworkGraphic = 2`

PNG Image converted from DDS Image

`[ResourceFileTypeFormat("json", "application/json")] JavaScriptObjectNotation = 3`

JavaScript Object Notation

`[ResourceFileTypeFormat("jpg", "image/jpeg")] JointPhotographicExpertsGroupImage = 4`

JPEG Image

`[ResourceFileTypeFormat("png", "image/png")] PortableNetworkGraphic = 5`

PNG Image

`[ResourceFileTypeFormat("jpg", "image/jpeg")] Ts4TranslucentJointPhotographicExpertsGroupImage = 7`

The Sims 4's proprietary image format in which a PNG alpha mask is embedded in the headers of a JPEG

```
[ResourceFileTypeFormat("xml", "application/xml")] TuningMarkup = 6
```

Tuning XML

```
[ResourceFileTypeFormat("yml", "application/x-yaml")] YetAnotherMarkupLanguage = 8
```

Yet Another Markup Language

Class ResourceFileTypeAttribute

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies the file type of a package resource type independent of storage within a DataBasePackedFile

```
[AttributeUsage(AttributeTargets.Field, AllowMultiple = false)]
public sealed class ResourceFileTypeAttribute : Attribute
```

Inheritance

object ← Attribute ← ResourceFileTypeAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) , Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) , Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) , Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) , Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) ,
object.ToString()

Constructors

ResourceFileTypeAttribute(ResourceFileType)

Specifies the file type of a package resource type independent of storage within a DataBasePackedFile

```
public ResourceFileTypeAttribute(ResourceFileType packageResourceFileType)
```

Parameters

`packageResourceFileType` ResourceFileType

Properties

ResourceFileType

Gets the file type of the decorated package resource type independent of storage within a DataBase PackedFile

```
public ResourceFileType ResourceFileType { get; }
```

Property Value

ResourceFileType

Class ResourceFileTypeFormatAttribute

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies the extension and mime type of a file type of a package resource independent of storage within a DataBasePackedFile

```
[AttributeUsage(AttributeTargets.Field, AllowMultiple = false)]
public sealed class ResourceFileTypeFormatAttribute : Attribute
```

Inheritance

object ← Attribute ← ResourceFileTypeFormatAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) , Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) , Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) , Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) , Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) ,
object.ToString()

Constructors

ResourceFileTypeFormatAttribute(string, string?)

Specifies the extension and mime type of a file type of a package resource independent of storage within a DataBasePackedFile

```
public ResourceFileTypeFormatAttribute(string extension, string? mimeType = null)
```

Parameters

`extension` string

`mimeType` string

Properties

Extension

Gets the extention used for files in the format

```
public string Extension { get; }
```

Property Value

string

MimeType

Gets the MIME type of the format

```
public string? MimeType { get; }
```

Property Value

string

Struct ResourceKey

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Represents the unique key of a resource within a DataBasePackedFile

```
public struct ResourceKey : IEquatable<ResourceKey>
```

Implements
IEquatable <ResourceKey>

Inherited Members
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Extension Methods
SmartSimUtilities.GetResourceKeyForLocale(ResourceKey, CultureInfo) ,
SmartSimUtilities.GetStringTableLocale(ResourceKey)

Constructors

ResourceKey(ResourceType, uint, ulong)

Represents the unique key of a resource within a DataBasePackedFile

```
public ResourceKey(ResourceType type, uint group, ulong fullInstance)
```

Parameters

`type` ResourceType

`group` uint

`fullInstance` ulong

Fields

Empty

Represents the empty ResourceKey

```
public static readonly ResourceKey Empty
```

Field Value

ResourceKey

FullInstance

A value that uniquely identifies an individual resource within a given type and group

```
public readonly ulong FullInstance
```

Field Value

ulong

Group

A value used to group related resources together

```
public readonly uint Group
```

Field Value

uint

Remarks

In many cases, this value can be 0, especially for custom content. However, Maxis content and some specific modding scenarios might use different group values to organize resources. Group values help in categorizing resources that belong to a specific set or context.

Type

The type of resource indicated by the key

```
public readonly ResourceType Type
```

Field Value

ResourceType

Properties

FullInstanceHex

Gets the hex representation of FullInstance

```
public readonly string FullInstanceHex { get; }
```

Property Value

string

FullTgi

Gets the full TGI or TGI ID (type, group, and instance in hex format) of the resource key

```
public readonly string FullTgi { get; }
```

Property Value

string

GroupHex

Gets the hex representation of Group

```
public readonly string GroupHex { get; }
```

Property Value

string

Instance

Gets the low order bits of FullInstance

```
public readonly uint Instance { get; }
```

Property Value

uint

InstanceEx

Gets the high order bits of FullInstance

```
public readonly uint InstanceEx { get; }
```

Property Value

uint

InstanceExHex

Gets the hex representation of InstanceEx

```
public readonly string InstanceExHex { get; }
```

Property Value

string

InstanceHex

Gets the hex representation of Instance

```
public readonly string InstanceHex { get; }
```

Property Value

string

TypeHex

Gets the hex representation of Type

```
public readonly string TypeHex { get; }
```

Property Value

string

Methods

Deconstruct(out ResourceType, out uint, out ulong)

Deconstructs the key into its components

```
public readonly void Deconstruct(out ResourceType type, out uint group, out
ulong fullInstance)
```

Parameters

`type` ResourceType

The Type

`group` uint

The Group

`fullInstance` ulong

The FullInstance

Equals(ResourceKey)

Indicates whether the current object is equal to another object of the same type.

```
public readonly bool Equals(ResourceKey other)
```

Parameters

`other` ResourceKey

An object to compare with this object.

Returns

bool

true if the current object is equal to the `other` parameter; otherwise, false.

Equals(object?)

Indicates whether this instance and a specified object are equal.

```
public override readonly bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current instance.

Returns

bool

true if `obj` and this instance are the same type and represent the same value; otherwise, false.

GetHashCode()

Returns the hash code for this instance.

```
public override readonly int GetHashCode()
```

Returns

int

A 32-bit signed integer that is the hash code for this instance.

Parse(string)

Parses a string into a ResourceKey

```
public static ResourceKey Parse(string s)
```

Parameters

`s` string

The string to parse

Returns

ResourceKey

Exceptions

FormatException

`s` is not in the correct format

Parse(string, IFormatProvider?)

Parses a string into a ResourceKey

```
public static ResourceKey Parse(string s, IFormatProvider? provider)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

Returns

ResourceKey

Exceptions

FormatException

`s` is not in the correct format

ToString()

Returns the fully qualified type name of this instance.

```
public override readonly string ToString()
```

Returns

string

The fully qualified type name.

TryParse(string?, out ResourceKey)

Tries to parse a string into a ResourceKey

```
public static bool TryParse(string? s, out ResourceKey result)
```

Parameters

`s` string

The string to parse

`result` ResourceKey

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

TryParse(string?, IFormatProvider?, out ResourceKey)

Tries to parse a string into a ResourceKey

```
public static bool TryParse(string? s, IFormatProvider? provider, out ResourceKey result)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

`result` ResourceKey

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

Operators

operator ==(ResourceKey, ResourceKey)

```
public static bool operator ==(ResourceKey left, ResourceKey right)
```

Parameters

`left` ResourceKey

`right` ResourceKey

Returns

bool

implicit operator ResourceKey(string)

Converts a string into a ResourceKey

```
[SuppressMessage("Usage", "CA2225:Operator overloads have named alternates", Justification =
"For convenience in Python.")]
public static implicit operator ResourceKey(string s)
```

Parameters

`s` string

The string to convert

Returns

ResourceKey

operator !=(ResourceKey, ResourceKey)

```
public static bool operator !=(ResourceKey left, ResourceKey right)
```

Parameters

`left` ResourceKey

`right` ResourceKey

Returns

bool

Enum ResourceKeyOrder

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies the order in which a ResourceKey sequence should be processed

```
public enum ResourceKeyOrder
```

Fields

`InstanceTypeGroup = 1`

The operation will process the ResourceKey sequence in the order FullInstance, then Type, then Group

`Preserve = 0`

The operation will process the ResourceKey sequence in the order in which it was originally encountered

`TypeGroupInstance = 2`

The operation will process the ResourceKey sequence in the order Type, then Group, then FullInstance

Class ResourceToolingMetadataAttribute

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Specifies that the ResourceType is tooling metadata and not intended for use by the game

```
[AttributeUsage(AttributeTargets.Field, AllowMultiple = false)]
public sealed class ResourceToolingMetadataAttribute : Attribute
```

Inheritance

object ← Attribute ← ResourceToolingMetadataAttribute

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) , Attribute.GetCustomAttributes(Assembly) ,
Attribute.GetCustomAttributes(Assembly, bool) , Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) , Attribute.GetCustomAttributes(MemberInfo) ,
Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) , Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object) ,
object.ToString()

Enum ResourceType

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

A type of resource within a DataBasePackedFile

```
[SuppressMessage("Design", "CA1028: Enum Storage should be Int32", Justification = "This is by nature of how the package files work. We have no control over that.")]
public enum ResourceType : uint
```

Fields

`AVI = 929579223`

AVI

`[ResourceFileType(ResourceFileType.TuningMarkup)] AccountRewardTuning = 3138329048`

Account Reward Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] AchievementCategoryTuning = 609337601`

Achievement Category Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] AchievementCollectionTuning = 80917605`

Achievement Collection Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] AchievementTuning = 2018877086`

Achievement Tuning

`[ResourceFileType(ResourceFileType.TuningMarkup)] ActionTuning = 209137191`

Action Tuning

`AgeGenderMap = 17805169`

Age Gender Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Ambience = 3537656749
```

Ambience

```
AnimationBoundaryConditionCache = 479834948
```

Animation Boundary Condition Cache

```
AnimationConstraintCache = 3794048034
```

Animation Constraint Cache

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AnimationStateMachine = 47570707
```

Animation State Machine

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AnimationTuning = 3994535597
```

Animation Tuning

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] ApartmentThumbnail1 = 2870590650
```

Apartment Thumbnail 1

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] ApartmentThumbnail2 = 3175683878
```

Apartment Thumbnail 2

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AspirationCategoryTuning =
3813727192
```

Aspiration Category Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AspirationTrackTuning =
3223387309
```

Aspiration Track Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AspirationTuning = 683034229
```

Aspiration Tuning

```
[ResourceFileType(ResourceFileType.Binary)] AudioConfiguration = 4244956094
```

Audio Configuration

```
AudioConfigurationReference = 968010314
```

Audio Configuration Reference

```
AudioEffects = 32437818
```

Audio Effects

```
AudioVocals = 27600859
```

Audio Vocals

```
AutomationTuning = 3671155384
```

Automation Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] AwayActionTuning = 2947394632
```

Away Action Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BalloonTuning = 3966406598
```

Balloon Tuning

```
BatchFixHistory = 1810979774
```

Batch Fix History

```
[ResourceFileType(ResourceFileType.Binary)] BlendGeometry = 108833297
```

Blend Geometry

```
Block = 127102176
```

Block

```
Blueprint = 958717478
```

Blueprint

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] BlueprintImage = 3543935006
```

Blueprint Image

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] BodyPartThumbnail = 1529359685
```

Body Part Thumbnail

```
BonePose = 55959718
```

Bone Pose

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BreedTuning = 874331941
```

Breed Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BroadcasterTuning = 3736796019
```

Broadcaster Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BucksPerkTuning = 3963461902
```

Bucks Perk Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BuffTuning = 1612179606
```

Buff Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BuildBuyTuning = 2919482169
```

Build Buy Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BusinessRuleTuning = 3102051436
```

Business Rule Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] BusinessTuning = 1977092083
```

Business Tuning

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] BuyBuildThumbnail = 1009419847
```

Buy Build Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASCameraTuning = 626712290
```

CAS Camera Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASLightingTuning = 2148152976
```

CAS Lighting Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASMenuItemTuning = 213537012
```

CAS Menu Item Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASMenuTuning = 2472182722
```

CAS Menu Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASModifierTuning = 4088135484
```

CAS Modifier Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASOccultSkintoneTuning =
2076404814
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CAS Occult Skintone Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASOccultTuning = 563571177
```

CAS Occult Tuning

```
[ResourceFileType(ResourceFileType.Binary)] CASPart = 55242443
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CAS Part

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] CASPartThumbnail = 1008398834
```

CAS Part Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASPreferenceCategoryTuning =
3456433227
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CAS Preference Category Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASPreferenceItemTuning =
3966303522
```

CAS Preference Item Tuning

```
[ResourceFileType(ResourceFileType.Binary)] CASPreset = 3936561885
```

CAS Preset

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASStoriesAnswerTuning =
2163289367
```

CAS Stories Answer Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASStoriesQuestionTuning = 52718493
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CAS Stories Question Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASStoriesTraitChooserTuning = 2376930633
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CAS Stories Trait Chooser Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASThumbnailPartTuning = 2932986028
```

CAS Thumbnail Part Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CASTuning = 3796587143
```

CAS Tuning

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CMRF = 1700995767
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CMRF

```
[ResourceFileType(ResourceFileType.TuningMarkup)] CallToActionTuning = 4114068192
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Call To Action Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] CameraTuning = 306800208
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Camera Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] CareerEventTuning = 2487354146
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Career Event Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] CareerGigTuning = 3436908253
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Career Gig Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] CareerLevelTuning = 745582072
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Career Level Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] CareerTrackTuning = 1221024995
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Career Track Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] CareerTuning = 1939434475
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Career Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] ClanTuning = 3737052837
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Clan Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] ClanValueTuning = 2576273579
```

Clan Value Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClientTutorialTuning = 645072749
```

Client Tutorial Tuning

```
[ResourceFileType(ResourceFileType.Binary)] Clip = 1797309683
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Clip

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ClipHeader = 3158986820
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Clip Header

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[ResourceFileType(ResourceFileType.TuningMarkup)] ClubInteractionGroupTuning =
4195351092
```

Club Interaction Group Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ClubSeedTuning = 794407991
```

Club Seed Tuning

```
ColorBlendedTerrain = 2934464740
```

Color Blended Terrain

```
ColorList = 2810271350
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Color List

```
ColorTimelineData = 4250380503
```

Color Timeline Data

```
Column = 493744591
```

Column

```
[ResourceFileType(ResourceFileType.Binary)] CombinedTuning = 1659456824
```

Combined Binary Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ConditionalLayerTuning =
2441338001
```

Conditional Layer Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Credits = 1721014748
```

Credits

```
Cursor = 647463969
```

Cursor

```
[ResourceFileType(ResourceFileType.Binary)] CutoutInfoTable = 2177505808
```

Cutout Info Table

```
[ResourceFileType(ResourceFileType.DirectDrawSurface)] DSTImage = 11720834
```

DST Image

```
DecoTrim = 332336850
```

Deco Trim

```
[ResourceFileType(ResourceFileType.Binary)] DeformerMap = 3678658665
```

Deformer Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DetectiveClueTuning = 1400130038
```

Detective Clue Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DevelopmentalMilestoneTuning =
3307360148
```

Developmental Milestone Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] DramaNodeTuning = 626258997
```

Drama Node Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] EnsembleTuning = 3112702240
```

Ensemble Tuning

```
Fence = 68746794
```

Fence

```
Floor = 3036111561
```

Floor

```
FloorTrim = 2227319321
```

Floor Trim

```
[ResourceFileType(ResourceFileType.TuningMarkup)] FontConfiguration = 53690476
```

Font Configuration

```
[ResourceFileType(ResourceFileType.Binary)] Footprint = 3548561239
```

Footprint

```
Footprint3 = 3073696847
```

Footprint 3

```
Foundation = 799971390
```

Foundation

```
FountainTrim = 3886917533
```

Fountain Trim

```
Frieze = 2690089244
```

Frieze

```
[ResourceFileType(ResourceFileType.TuningMarkup)] GameRulesetTuning = 3779558936
```

Game Ruleset Tuning

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GenericMTX = 2885921078
```

Generic MTX

```
[ResourceFileType(ResourceFileType.Binary)] Geometry = 22681673
```

Geometry

```
[ResourceFileType(ResourceFileType.TuningMarkup)] GuidanceTipTuning = 3567295165
```

Guidance Tip Tuning

```
HalfWall = 2438063804
```

HalfWall

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HeadlineTuning = 4093714525
```

Headline Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HolidayDefinitionTuning =
238120813
```

Holiday Definition Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HolidayTraditionTuning =
1070408838
```

Holiday Tradition Tuning

```
HouseholdDescription = 1923050575
```

Household Description

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HouseholdMilestoneTuning =
963831539
```

Household Milestone Tuning

```
HouseholdTemplate = 3015981296
```

Household Template

```
[ResourceFileType(ResourceFileType.TuningMarkup)] HsvTweakerSettingsTuning =
603754750
```

Hsv Tweaker Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] InteractionTuning = 3900887599
```

Interaction Tuning

```
LRLEImage = 734023391
```

LRLE Image

```
Ladder = 3423141965
```

Ladder

```
Light = 62178845
```

Light

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LocomotionBuilder = 87703163
```

Locomotion Builder

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LocomotionConfig = 2600355829
```

Locomotion Config

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotDecorationPresetTuning =
3726571771
```

Lot Decoration Preset Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotDecorationTuning = 4264407467
```

Lot Decoration Tuning

```
[ResourceFileType(ResourceFileType.Binary)] LotDescription = 26488364
```

Lot Description

```
LotFootprintReference = 3221768598
```

Lot Footprint Reference

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] LotPreviewThumbnail = 221481530
```

Lot Preview Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotTuning = 3632270694
```

Lot Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LotTypeEventMap = 305120874
```

Lot Type Event Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] LunarCycleTuning = 1430862616
```

Lunar Cycle Tuning

```
MTXCatalog = 1845434982
```

MTX Catalog

```
Magalog = 3105757403
```

Magalog

```
MagazineCollection = 1946487583
```

Magazine Collection

```
MaterialDefinition = 30467933
```

Material Definition

```
MaterialSet = 33659250
```

Material Set

```
MaxisWorldPipeline1 = 3773657385
```

Maxis World Pipeline 1

```
MaxisWorldPipeline2 = 4196775902
```

Maxis World Pipeline 2

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] MemorialThumbnail = 21
```

Memorial Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ModalMusicMapping = 444925637
```

Modal Music Mapping

```
Model = 23466547
```

Model

```
ModelCutout = 123169303
```

Model Cutout

```
ModelLOD = 30478132
```

Model LOD

```
ModularPiece = 2568481485
```

Modular Piece

```
ModularPieceCatalog = 2688883901
```

Modular Piece Catalog

```
[ResourceFileType(ResourceFileType.TuningMarkup)] MoodTuning = 3128647864
```

Mood Tuning

```
MusicData = 3254962032
```

Music Data

```
NameMap = 23462796
```

Name Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NarrativeTuning = 1047870521
```

Narrative Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NativeBuildBuyTuning = 445966772
```

Native Build Buy Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NativeSeasonsWeatherTuning =
3023723217
```

Native Seasons Weather Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] NotebookEntryTuning = 2567109238
```

Notebook Entry Tuning

```
[ResourceFileType(ResourceFileType.Binary)] ObjectCatalog = 832458525
```

Object Catalog

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ObjectCatalogSet = 4283826444
```

Object Catalog Set

```
[ResourceFileType(ResourceFileType.Binary)] ObjectDefinition = 3235601127
```

Object Definition

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectModifiers = 3794908120
```

Object Modifiers

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectPartTuning = 1900520272
```

Object Part Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectStateTuning = 1526890910
```

Object State Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectTuning = 3055412916
```

Object Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ObjectiveTuning = 6899006
```

Objective Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] OpenStreetDirectorTuning =
1265622724
```

Open Street Director Tuning

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OpenTypeFont = 628714954
```

OpenType Font

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] PNG = 18461549
```

PNG

```
[ResourceFileType(ResourceFileType.DirectDrawSurfaceAsPortableNetworkGraphic)]
PNGImage = 796721156
```

PNG Image converted from DDS Image

```
[ResourceFileType(ResourceFileType.DirectDrawSurfaceAsPortableNetworkGraphic)]
PNGImage2 = 796721158
```

PNG Image converted from DDS Image 2

```
Path = 975082574
```

Path

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] PetBreedCoatPatternThumbnail = 2389771869
```

Pet Breed Coat Pattern Thumbnail

```
PetCoatBrush = 2642310488
```

Pet Coat Brush

```
PetCoatPattern = 3302993517
```

Pet Coat Pattern

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] PetFacePresetThumbnail = 3061216162
```

Pet Face Preset Thumbnail

```
PetPeltLayer = 649036600
```

Pet Pelt Layer

```
[ResourceFileType(ResourceFileType.TuningMarkup)] PieMenuCategoryTuning = 65657188
```

Pie Menu Category Tuning

```
Platform = 2579526276
```

Platform

```
Pond = 1342387004
```

Pond

```
PoolTrim = 2782919923
```

Pool Trim

```
[ResourceFileType(ResourceFileType.TuningMarkup)] PostureTuning = 2909789983
```

Posture Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] PubertyTuning = 1209600077
```

Puberty Tuning

```
QueryableWorldMaskManifest = 1316074726
```

Queryable World Mask Manifest

```
[ResourceFileType(ResourceFileType.DirectDrawSurface)] RLE2Image = 877907861
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RLE 2 Image

```
[ResourceFileType(ResourceFileType.DirectDrawSurface)] RLESImage = 3129306232
```

RLES Image

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RabbitHoleTuning = 2976568058
```

Rabbit Hole Tuning

```
Railing = 471658999
```

Railing

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RecipeTuning = 3952605219
```

Recipe Tuning

```
RecordBundle = 1051371644
```

Record Bundle

```
[ResourceFileType(ResourceFileType.Binary)] RegionDescription = 3596464121
```

Region Description

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[ResourceFileType(ResourceFileType.Binary)] RegionMap = 2887187436
```

Region Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RegionSortTuning = 1062717573
```

Region Sort Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RegionTuning = 1374134669
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Region Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RelationshipBitTuning = 151314192
```

Relationship Bit Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RelationshipLockTuning =
2922702451
```

Relationship Lock Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererCensorTuning = 3788962514
```

Renderer Censor Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererFadeTuning = 69178842
```

Renderer Fade Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalDOFSettingsTuning =
2368970974
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Renderer Global DOF Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)]
RendererGlobalHighlightSettingsTuning = 3162386069
```

Renderer Global Highlight Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalLightSettingsTuning
= 1790074981
```

Renderer Global Light Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalSSAOSettingsTuning =
795796239
```

Renderer Global SSAO Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RendererGlobalShadowSettingsTuning
= 1160337457
```

Renderer Global Shadow Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)]
RendererGlobalVolLightScatteringSettingsTuning = 3319940574
```

Renderer Global Vol Light Scattering Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RewardTuning = 1873057832
```

Reward Tuning

```
Rig = 2393838558
```

Rig

```
RoadDefinition = 2422433294
```

Road Definition

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RoleStateTuning = 239932923
```

Role State Tuning

```
Roof = 2448276798
```

Roof

```
RoofPattern = 4058889606
```

Roof Pattern

```
RoofTrim = 2956008719
```

Roof Trim

```
[ResourceFileType(ResourceFileType.TuningMarkup)] RoyaltyTuning = 938421991
```

Royalty Tuning

```
S4sMergedPackageManifest = 2142678410
```

Sims 4 Studio Merged Package Manifest

```
[ResourceFileType(ResourceFileType.JointPhotographicExpertsGroupImage)]
SaveGameCustomTexture = 3901600607
```

Save Game Custom Texture

```
[ResourceFileType(ResourceFileType.Binary)] SaveGameData = 13
```

Save Game Data

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameHouseholdThumbnail = 20
```

Save Game Household Thumbnail

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotLevelThumbnail = 18
```

Save Game Lot Level Thumbnail

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotLevelThumbnailMask = 19
```

Save Game Lot Level Thumbnail Mask

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotThumbnail1 = 15
```

Save Game Lot Thumbnail 1

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameLotThumbnail2 = 16
```

Save Game Lot Thumbnail 2

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SaveGameSimCustomTexture = 4175512954
```

Save Game Sim Custom Texture

```
[ResourceFileType(ResourceFileType.Binary)] SaveSpecificRelationalDataStorage =
1382263428
```

Save-Specific Relational Data Storage

This is a ZIP archive's octet stream set to store, not any level of compression. Use DBPF ZLIB compression. Each entry should be a SQLite database file using the `.sqlite` extension, the name of

which is a valid UUID.

```
ScaleFormGFX = 1659684250
```

ScaleForm GFX

```
Sculpt = 2635774068
```

Sculpt

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SeasonTuning = 3381515358
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Season Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ServiceNPCTuning = 2629964386
```

Service NPC Tuning

```
[ResourceFileType(ResourceFileType.Binary)] ShaderPrecomp = 670978917
```

Game Shaders for Windows

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SicknessTuning = 3288062174
```

Sickness Tuning

```
[ResourceFileType(ResourceFileType.Binary)] SimData = 1415235194
```

Sim Data

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SimFeaturedOutfitThumbnail = 3449676359
```

Sim Featured Outfit Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SimFilterTuning = 1846401695
```

Sim Filter Tuning

```
SimHotspotControl = 2333671278
```

Sim Hotspot Control

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SimHouseholdThumbnail = 1003770887
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Sim Household Thumbnail

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SimInfo = 39769844
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Sim Info

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SimInfoFixupTuning = 3797424274
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Sim Info Fixup Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SimLightingTuning = 2648399834
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Sim Lighting Tuning

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SimModifier = 3321263678
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Sim Modifier

```
SimPreset = 273810874
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Sim Preset

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[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] SimPresetThumbnail = 2626836499
```

Sim Preset Thumbnail

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[ResourceFileType(ResourceFileType.TuningMarkup)] SimTemplateTuning = 212125579
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Sim Template Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SituationGoalSetTuning =
2649944562
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Situation Goal Set Tuning

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Situation Goal Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SituationJobTuning = 2617738591
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Situation Job Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SituationTuning = 4223905515
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Situation Tuning

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[ResourceFileType(ResourceFileType.Binary)] Skintone = 55867754
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Skintone

```
[ResourceFileType(ResourceFileType.Binary)] SkyBoxTextureData = 1906592201
```

Sky Box Texture Data

```
[ResourceFileType(ResourceFileType.Binary)] Slot = 3540272417
```

Slot

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[ResourceFileType(ResourceFileType.TuningMarkup)] SlotTypeSetTuning = 1058419973
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Slot Type Set Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SlotTypeTuning = 1772477092
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Slot Type Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SnippetTuning = 2113017500
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Snippet Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SocialGroupTuning = 776446212
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Social Group Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] SoundMixProperties = 1091959253
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Sound Mix Properties

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[ResourceFileType(ResourceFileType.TuningMarkup)] SoundModifierMapping = 2776023783
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Sound Modifier Mapping

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[ResourceFileType(ResourceFileType.TuningMarkup)] SoundProperties = 455450660
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Sound Properties

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Spandrel = 1057772186
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Spandrel

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Spawner = 1220708729
```

Spawner

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[ResourceFileType(ResourceFileType.TuningMarkup)] SpellTuning = 523506649
```

Spell Tuning

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Stairs = 2585840924
```

Stairs

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StaticCommodityTuning =
1359443523
```

Static Commodity Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StatisticTuning = 865846717
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Statistic Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StoryArcTuning = 1613438381
```

Story Arc Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StoryChapterTuning = 1250314810
```

Story Chapter Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StrategySetTuning = 1646578134
```

Strategy Set Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] StreetTuning = 4142189312
```

Street Tuning

```
[ResourceFileType(ResourceFileType.Binary)] StringTable = 570775514
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String Table

```
Style = 2673671952
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Style

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StyledLook = 1908258978
```

Styled Look

```
[ResourceFileType(ResourceFileType.TuningMarkup)] SubrootTuning = 3086978965
```

Subroot Tuning

```
SyncPointSchema = 757559827
```

Sync Point Schema

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TagCategoriesMetadataTuning =
470996056
```

Tag Categories Metadata Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TagSetTuning = 1228493570
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Tag Set Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] TagTraitGroupMetadataTuning =
2302558876
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Tag Trait Group Metadata Tuning

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[ResourceFileType(ResourceFileType.TuningMarkup)] TagTuning = 3708124620
```

Tag Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TagsMetadataTuning = 2646182397
```

Tags Metadata Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)]
TelemetryMemoryUsageTelemetrySettingsTuning = 2312527733
```

Telemetry Memory Usage Telemetry Settings Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TelemetryTuning = 2900743302
```

Telemetry Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TemplateChooserTuning =
1220728301
```

Template Chooser Tuning

```
[ResourceFileType(ResourceFileType.Binary)] TerrainBlendMap = 1032205008
```

Terrain Blend Map

```
[ResourceFileType(ResourceFileType.Binary)] TerrainData = 2422433293
```

Terrain Data

[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)] TerrainHeightMap = 718378482

Terrain Height Map

[ResourceFileType(ResourceFileType.Binary)] TerrainKDTree = 54209382

Terrain KD Tree

[ResourceFileType(ResourceFileType.Binary)] TerrainMesh = 2922985887

Terrain Mesh

TerrainPaint = 3955994988

Terrain Paint

TerrainSizeInfo = 2422361115

Terrain Size Info

TerrainTool = 338149641

Terrain Tool

[ResourceFileType(ResourceFileType.TuningMarkup)] TestBasedScoreTuning = 1332976878

Test Based Score Tuning

TheSimsResourceWorkshopProprietary = 20281636

A proprietary resource used by The Sims Resource Workshop

[ResourceFileType(ResourceFileType.TuningMarkup)] ThriftStoreTuning = 3841838852

Thrift Store Tuning

[ResourceFileType(ResourceFileType.Binary)] ThumbnailCache = 3107625237

Thumbnail Cache

ThumbnailExtra1 = 382364612

Thumbnail Extra 1

ThumbnailExtra2 = 2953939989

Thumbnail Extra 2

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ThumbnailTuning = 3787619543
```

Thumbnail Tuning

```
TimelineEvents = 111944783
```

Timeline Events

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TopicTuning = 1938713686
```

Topic Tuning

```
TrackMask = 53633251
```

Track Mask

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TraitTuning = 3412057543
```

Trait Tuning

```
TrayItem = 713711138
```

Tray Item

```
[ResourceFileType(ResourceFileType.Binary)] Trim = 1992095756
```

Trim

```
TrueTypeFont = 661431481
```

TrueType Font

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Tuning = 62078431
```

Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TutorialTipTuning = 2410930353
```

Tutorial Tip Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] TutorialTuning = 3762955427
```

Tutorial Tuning

```
UIControlEventMap = 3185058337
```

UI Control Event Map

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UIEventModeMapping = 2581168265
```

UI Event Mode Mapping

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityCourseDataTuning = 689745854
```

University Course Data Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityCourseMajorTuning = 660124491
```

University Course Major Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UniversityTuning = 3646477745
```

University Tuning

```
UnknownWorld2 = 398508673
```

Unknown World 2

```
UnknownWorld3 = 2528165143
```

Unknown World 3

```
UnknownWorld5 = 3162565977
```

Unknown World 5

```
Unspecified = 0
```

Unspecified

```
[ResourceFileType(ResourceFileType.TuningMarkup)] UserInterfaceInfoTuning = 3099531875
```

User Interface Info Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VenueTuning = 3871070174
```

Venue Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VideoGlobalTuning = 2441613679
```

Video Global Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VideoPlaylistTuning = 1086277067
```

Video Playlist Tuning

```
VisualEffects = 454631497
```

Visual Effects

```
VisualEffectsInstanceMap = 454631498
```

Visual Effects Instance Map

```
VisualEffectsMerged = 3931183280
```

Visual Effects Merged

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VoiceEffect = 1942696642
```

Voice Effect

```
[ResourceFileType(ResourceFileType.TuningMarkup)] VoicePlugin = 3313685243
```

Voice Plugin

```
WMRFReference = 551031958
```

WMRF Reference

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WalkByTuning = 1070998590
```

Walk By Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] Walkstyle = 666901909
```

Walkstyle

```
Wall = 3589339425
```

Wall

```
WallTrim = 4264756878
```

Wall Trim

```
WaterManifest = 386491005
```

Water Manifest

```
WaterMask = 1207820220
```

Water Mask

```
WaterMaskList = 2763687493
```

Water Mask List

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WeatherEventTuning = 1476851130
```

Weather Event Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WeatherForecastTuning =
1233072753
```

Weather Forecast Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] WhimTuning = 1956251190
```

Whim Tuning

```
WindowSet = 2834806039
```

Window Set

```
WorldCameraInfo = 2301381514
```

World Camera Info

```
WorldCameraMesh = 4211986434
```

World Camera Mesh

```
[ResourceFileType(ResourceFileType.Binary)] WorldConditionalData = 3494881377
```

World Conditional Data

```
[ResourceFileType(ResourceFileType.Binary)] WorldData = 2164920365
```

World Data

```
[ResourceFileType(ResourceFileType.Binary)] WorldDescription = 2793466443
```

World Description

```
WorldFileHeader = 4033034633
```

World File Header

```
WorldLandingStrip = 1332898750
```

World Landing Strip

```
WorldLightsInfo = 418629235
```

World Lights Info

```
WorldLotArchitecture = 311764532
```

World Lot Architecture

```
WorldLotObjects = 2438369240
```

World Lot Objects

```
WorldLotParameterInfo = 1006173574
```

World Lot ParameterInfo

```
WorldManifest = 2026421476
```

World Manifest

```
[ResourceFileType(ResourceFileType.Binary)] WorldMap = 482361971
```

World Map

```
[ResourceFileType(ResourceFileType.Binary)] WorldObjectData = 4239499748
```

World Object Data

```
WorldOffLotMesh = 170032079
```

World Off Lot Mesh

```
WorldRoadPolys = 1866496362
```

World Road Polys

```
[ResourceFileType(ResourceFileType.Binary)] WorldTimelineColor = 422580512
```

World Timeline Color

```
WorldVisualEffectsInfo = 1541576451
```

World Visual Effects Info

```
WorldWaterManifest = 356327961
```

World Water Manifest

```
[ResourceFileType(ResourceFileType.Ts4TranslucentJointPhotographicExpertsGroupImage)
] WorldmapLotThumbnail = 2717855684
```

Worldmap Lot Thumbnail

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ZoneDirectorTuning = 4183335058
```

Zone Director Tuning

```
[ResourceFileType(ResourceFileType.TuningMarkup)] ZoneModifierTuning = 1008568217
```

Zone Modifier Tuning

```
ZoneObjectData = 6
```

Zone Object Data

Class SmartSimUtilities

Namespace: LlamaLogic.Packages

Assembly: LlamaLogic.Packages.dll

Provides utility methods specific to The Sims 4

```
public static class SmartSimUtilities
```

Inheritance

object ← SmartSimUtilities

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetResourceKeyForLocale(ResourceKey, CultureInfo)

Gets the ResourceKey for this string table resource with the specified `locale`

```
public static ResourceKey GetResourceKeyForLocale(this ResourceKey key, CultureInfo locale)
```

Parameters

`key` ResourceKey

`locale` CultureInfo

Returns

ResourceKey

Exceptions

ArgumentOutOfRangeException

The locale is not supported by The Sims 4

GetStringTableLocale(ResourceKey)

Gets the locale of a string table resource in The Sims 4 based on its `key`

```
public static CultureInfo GetStringTableLocale(this ResourceKey key)
```

Parameters

`key` ResourceKey

Returns

CultureInfo

Exceptions

ArgumentOutOfRangeException

The high order byte of the instance does not correspond to a locale supported by The Sims 4

GetStringTableResourceKey(CultureInfo, uint, ulong)

Gets the ResourceKey of a string table resource in The Sims 4 based on its `locale`, `group`, and `fullInstance`

```
public static ResourceKey GetStringTableResourceKey(this CultureInfo locale, uint group,
ulong fullInstance)
```

Parameters

`locale` CultureInfo

`group` uint

`fullInstance` ulong

Returns

ResourceKey

Exceptions

ArgumentOutOfRangeException

The locale is not supported by The Sims 4

Namespace LlamaLogic.Packages.Cryptography

Classes

Fnv24

Represents the 24-bit implementation of the Fowler–Noll–Vo hash algorithm

Fnv32

Represents the 32-bit implementation of the Fowler–Noll–Vo hash algorithm

Fnv56

Represents the 56-bit implementation of the Fowler–Noll–Vo hash algorithm

Fnv64

Represents the 64-bit implementation of the Fowler–Noll–Vo hash algorithm

FnvHash

Represents the base class from which this library's implementations of the Fowler–Noll–Vo hash algorithms derive

Class Fnv24

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 24-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public sealed class Fnv24 : Fnv32, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv32 ← Fnv24

Implements

ICryptoTransform, IDisposable

Inherited Members

Fnv32.HashSize , FnvHash.ComputeHash(string) , FnvHash.Initialize() , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly uint EmptyStringHash
```

Field Value

uint

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static uint GetHash(string? text)
```

Parameters

`text` string

Returns

uint

SetHighBit(uint)

Sets the high bit of a 24-bit Fowler–Noll–Vo hash

```
public static uint SetHighBit(uint hash)
```

Parameters

`hash` uint

Returns

uint

Class Fnv32

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 32-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public class Fnv32 : FnvHash, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv32

Implements

ICryptoTransform, IDisposable

Derived

Fnv24

Inherited Members

FnvHash.hash , FnvHash.ComputeHash(string) , FnvHash.HashCore(byte[], int, int) , FnvHash.HashFinal() , FnvHash.Initialize() , HashAlgorithm.HashSizeValue , HashAlgorithm.HashValue , HashAlgorithm.State , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.Dispose(bool) , HashAlgorithm.HashCore(ReadOnlySpan<byte>) , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.TryHashFinal(Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Fnv32()

Initializes a new instance of the Fnv32 class

```
public Fnv32()
```

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly uint EmptyStringHash
```

Field Value

uint

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

HashSize

Gets the size, in bits, of the computed hash code.

```
public override int HashSize { get; }
```

Property Value

int

The size, in bits, of the computed hash code.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static uint GetHash(string? text)
```

Parameters

`text` string

Returns

uint

SetHighBit(uint)

Sets the high bit of a 32-bit Fowler–Noll–Vo hash

```
public static uint SetHighBit(uint hash)
```

Parameters

`hash` uint

Returns

uint

Class Fnv56

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 56-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public sealed class Fnv56 : Fnv64, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv64 ← Fnv56

Implements

ICryptoTransform, IDisposable

Inherited Members

Fnv64.HashSize , FnvHash.ComputeHash(string) , FnvHash.Initialize() , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly ulong EmptyStringHash
```

Field Value

ulong

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static ulong GetHash(string? text)
```

Parameters

`text` string

Returns

ulong

SetHighBit(ulong)

Sets the high bit of a 56-bit Fowler–Noll–Vo hash

```
public static ulong SetHighBit(ulong hash)
```

Parameters

`hash` ulong

Returns

ulong

Class Fnv64

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the 64-bit implementation of the Fowler–Noll–Vo hash algorithm

```
public class Fnv64 : FnvHash, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash ← Fnv64

Implements

ICryptoTransform, IDisposable

Derived

Fnv56

Inherited Members

FnvHash.hash , FnvHash.ComputeHash(string) , FnvHash.HashCore(byte[], int, int) , FnvHash.HashFinal() , FnvHash.Initialize() , HashAlgorithm.HashSizeValue , HashAlgorithm.HashValue , HashAlgorithm.State , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.Dispose(bool) , HashAlgorithm.HashCore(ReadOnlySpan<byte>) , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.TryHashFinal(Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Fnv64()

Initializes a new instance of the Fnv64 class

```
public Fnv64()
```

Fields

EmptyStringHash

Gets the hash value for an empty string

```
public static readonly ulong EmptyStringHash
```

Field Value

ulong

Properties

Hash

Gets the value of the computed hash code.

```
public override byte[]? Hash { get; }
```

Property Value

byte[]

The current value of the computed hash code.

Exceptions

CryptographicUnexpectedOperationException

HashValue is null.

ObjectDisposedException

The object has already been disposed.

HashSize

Gets the size, in bits, of the computed hash code.

```
public override int HashSize { get; }
```

Property Value

int

The size, in bits, of the computed hash code.

Methods

GetHash(string?)

Computes the hash value for the specified string

```
public static ulong GetHash(string? text)
```

Parameters

`text` string

Returns

ulong

SetHighBit(ulong)

Sets the high bit of a 64-bit Fowler–Noll–Vo hash

```
public static ulong SetHighBit(ulong hash)
```

Parameters

`hash` ulong

Returns

ulong

Class FnvHash

Namespace: LlamaLogic.Packages.Cryptography

Assembly: LlamaLogic.Packages.dll

Represents the base class from which this library's implementations of the Fowler–Noll–Vo hash algorithms derive

```
public abstract class FnvHash : HashAlgorithm, ICryptoTransform, IDisposable
```

Inheritance

object ← HashAlgorithm ← FnvHash

Implements

ICryptoTransform, IDisposable

Derived

Fnv32, Fnv64

Inherited Members

HashAlgorithm.HashSizeValue , HashAlgorithm.HashValue , HashAlgorithm.State , HashAlgorithm.Clear() , HashAlgorithm.ComputeHash(byte[]) , HashAlgorithm.ComputeHash(byte[], int, int) , HashAlgorithm.ComputeHash(Stream) , HashAlgorithm.ComputeHashAsync(Stream, CancellationToken) , HashAlgorithm.Create() , HashAlgorithm.Create(string) , HashAlgorithm.Dispose() , HashAlgorithm.Dispose(bool) , HashAlgorithm.HashCore(ReadOnlySpan<byte>) , HashAlgorithm.TransformBlock(byte[], int, int, byte[], int) , HashAlgorithm.TransformFinalBlock(byte[], int, int) , HashAlgorithm.TryComputeHash(ReadOnlySpan<byte>, Span<byte>, out int) , HashAlgorithm.TryHashFinal(Span<byte>, out int) , HashAlgorithm.CanReuseTransform , HashAlgorithm.CanTransformMultipleBlocks , HashAlgorithm.Hash , HashAlgorithm.HashSize , HashAlgorithm.InputBlockSize , HashAlgorithm.OutputBlockSize , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

FnvHash(ulong, ulong)

Represents the base class from which this library's implementations of the Fowler–Noll–Vo hash algorithms derive

```
protected FnvHash(ulong prime, ulong offset)
```

Parameters

`prime` ulong

`offset` ulong

Fields

hash

The hash aggregate

```
protected ulong hash
```

Field Value

ulong

Methods

ComputeHash(string?)

Computes the hash value for the specified string

```
[SuppressMessage("Globalization", "CA1308: Normalize strings to uppercase", Justification =
"It's not our fault Maxis did this wrong, Code Analyzer.")]
public byte[] ComputeHash(string? value)
```

Parameters

`value` string

Returns

byte[]

HashCore(byte[], int, int)

When overridden in a derived class, routes data written to the object into the hash algorithm for computing the hash.

```
protected override void HashCore(byte[] array, int ibStart, int cbSize)
```

Parameters

`array` byte[]

The input to compute the hash code for.

`ibStart` int

The offset into the byte array from which to begin using data.

`cbSize` int

The number of bytes in the byte array to use as data.

HashFinal()

When overridden in a derived class, finalizes the hash computation after the last data is processed by the cryptographic hash algorithm.

```
protected override byte[] HashFinal()
```

Returns

byte[]

The computed hash code.

Initialize()

Resets the hash algorithm to its initial state.

```
public override void Initialize()
```

Namespace LlamaLogic.Packages.Formats

Classes

DirectDrawSurface

Provides methods for dealing with diffuse surface textures in Direct Draw Surfaces

Ts4TranslucentJointPhotographicExpertsGroupImage

Provides methods for dealing with Maxis Translucent JPEGs

Class DirectDrawSurface

Namespace: LlamaLogic.Packages.Formats

Assembly: LlamaLogic.Packages.dll

Provides methods for dealing with diffuse surface textures in Direct Draw Surfaces

```
public static class DirectDrawSurface
```

Inheritance

object ← DirectDrawSurface

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Adapted from S4S by Andrew

Methods

GetDstDataFromPngData(ReadOnlyMemory<byte>)

Convert a Portable Network Graphic to a shuffled DirectDraw Surface compatible with The Sims 4

```
public static ReadOnlyMemory<byte> GetDstDataFromPngData(ReadOnlyMemory<byte> pngData)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data of the Portable Network Graphic

Returns

ReadOnlyMemory<byte>

GetDstDataFromPngDataAsync(ReadOnlyMemory<byte>, CancellationToken)

Convert a Portable Network Graphic to a shuffled DirectDraw Surface compatible with The Sims 4, asynchronously

```
public static Task<ReadOnlyMemory<byte>> GetDstDataFromPngDataAsync(ReadOnlyMemory<byte>
pngData, CancellationToken cancellationToken = default)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data of the Portable Network Graphic

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

GetPngDataFromDdsData(ReadOnlyMemory<byte>)

Convert a DirectDraw Surface to a Portable Network Graphic

```
public static ReadOnlyMemory<byte> GetPngDataFromDdsData(ReadOnlyMemory<byte> ddsData)
```

Parameters

`ddsData` ReadOnlyMemory<byte>

The data of a DirectDraw Surface

Returns

ReadOnlyMemory<byte>

GetPngDataFromDdsDataAsync(ReadOnlyMemory<byte>, CancellationToken)

Convert a DirectDraw Surface to a Portable Network Graphic, asynchronously

```
public static Task<ReadOnlyMemory<byte>> GetPngDataFromDdsDataAsync(ReadOnlyMemory<byte>
ddsData, CancellationToken cancellationToken = default)
```

Parameters

`ddsData` ReadOnlyMemory<byte>

The data of a DirectDraw Surface

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

Class Ts4TranslucentJointPhotographicExpertsGroupImage

Namespace: LlamaLogic.Packages.Formats

Assembly: LlamaLogic.Packages.dll

Provides methods for dealing with Maxis Translucent JPEGs

```
public static class Ts4TranslucentJointPhotographicExpertsGroupImage
```

Inheritance

object ← Ts4TranslucentJointPhotographicExpertsGroupImage

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ConvertPngToTranslucentJpeg(ReadOnlyMemory<byte>)

Converts the data for a PNG the data for a Maxis Translucent JPEG

```
public static ReadOnlyMemory<byte> ConvertPngToTranslucentJpeg(ReadOnlyMemory<byte> pngData)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data for a PNG

Returns

ReadOnlyMemory<byte>

ConvertPngToTranslucentJpegAsync(ReadOnlyMemory<byte>, CancellationToken)

Converts the data for a PNG the data for a Maxis Translucent JPEG asynhronously

```
public static Task<ReadOnlyMemory<byte>>
ConvertPngToTranslucentJpegAsync(ReadOnlyMemory<byte> pngData,
CancellationToken cancellationToken)
```

Parameters

`pngData` ReadOnlyMemory<byte>

The data for a PNG

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

ConvertTranslucentJpegToPng(ReadOnlyMemory<byte>)

Converts the data for a Maxis Translucent JPEG the data for a PNG

```
public static ReadOnlyMemory<byte> ConvertTranslucentJpegToPng(ReadOnlyMemory<byte>
translucentJpegData)
```

Parameters

`translucentJpegData` ReadOnlyMemory<byte>

The data for a Maxis Translucent JPEG

Returns

ReadOnlyMemory<byte>

ConvertTranslucentJpegToPngAsync(ReadOnlyMemory<byte>, CancellationToken)

Converts the data for a Maxis Translucent JPEG the data for a PNG asynhronously

```
public static Task<ReadOnlyMemory<byte>>
ConvertTranslucentJpegToPngAsync(ReadOnlyMemory<byte> translucentJpegData, CancellationToken
cancellationToken = default)
```

Parameters

`translucentJpegData` ReadOnlyMemory<byte>

The data for a Maxis Translucent JPEG

`cancellationToken` CancellationToken

The token to monitor for cancellation requests

Returns

Task<ReadOnlyMemory<byte>>

Namespace LlamaLogic.Packages.Models

Classes

Model

Represents a model for a raw resource

StringTableModel

Represents a StringTable resource (🔒)

UnexpectedFormatDecodingException

Represents an exception thrown by an IModel<TSelf> when it attempts to decode a resource that is not in the expected format

Interfaces

IModel

Represents a model for a resource

IModel<TSelf>

Represents a model for a raw resource

Interface IModel

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a model for a resource

```
public interface IModel
```

Properties

ResourceName

Gets the name of this resource if it has one

```
string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static abstract ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static abstract string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static abstract Task<string?> GetNameAsync(Stream stream, CancellationToken
cancellationToken = default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Interface IModel<TSelf>

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a model for a raw resource

```
public interface IModel<TSelf> : IModel where TSelf : IModel<TSelf>
```

Type Parameters

`TSelf`

Inherited Members
IModel.SupportedTypes , IModel.GetName(Stream) , IModel.GetNameAsync(Stream, CancellationToken) , IModel.ResourceName , IModel.Encode()

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔁 🏃)

```
public static abstract TSelf Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

TSelf

Class Model

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a model for a raw resource

```
public abstract class Model : IModel
```

Inheritance

object ← Model

Implements

IModel

Derived

DataModel, GlobalModsManifestModel, ModFileManifestModel, StringTableModel

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ResourceName

Gets the name of this resource if it has one

```
public abstract string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet <ResourceType>

Methods

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public abstract ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory <byte>

GetName(ResourceType, Stream)

Gets the name of a resource from its raw data, if it has one

```
public static string? GetName(ResourceType type, Stream stream)
```

Parameters

`type` ResourceType

`stream` Stream

Returns

string

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(ResourceType, Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one

```
public static Task<string?> GetNameAsync(ResourceType type, Stream stream, CancellationToken
cancellationToken = default)
```

Parameters

`type` ResourceType

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task <string>

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Class StringTableModel

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents a StringTable resource (🔒)

```
public class StringTableModel : Model, IModel<StringTableModel>, IModel
```

Inheritance

object ← Model ← StringTableModel

Implements

IModel<StringTableModel>, IModel

Inherited Members

Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Count

Gets the number of entries in the string table

```
public int Count { get; }
```

Property Value

int

this[uint]

Gets/sets the string associated with a `keyHash`

```
public string this[uint keyHash] { get; set; }
```

Parameters

`keyHash` uint

Property Value

string

KeyHashes

Gets the key hashes in the string table

```
public IReadOnlyList<uint> KeyHashes { get; }
```

Property Value

IReadOnlyList<uint>

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

AddNew(string)

Adds `string` to the string table, returning its key hash

```
public uint AddNew(string @string)
```

Parameters

`string` string

Returns

uint

Exceptions

ArgumentException

The key hash generated for `string` already existed in the string table

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔄 🏃)

```
public static StringTableModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

StringTableModel

Delete(uint)

Deletes the string associated with a `keyHash`, returning true if the key hash existed in the string table

```
public bool Delete(uint keyHash)
```

Parameters

`keyHash` uint

Returns

bool

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

Get(uint)

Gets the string associated with a `keyHash`

```
public string Get(uint keyHash)
```

Parameters

`keyHash` uint

Returns

string

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task <string>

Set(uint, string)

Sets the `string` associated with a `keyHash`, returning true if the key hash did not already exist in the string table

```
public bool Set(uint keyHash, string @string)
```

Parameters

`keyHash` uint

`string` string

Returns

bool

Class UnexpectedFormatDecodingException

Namespace: LlamaLogic.Packages.Models

Assembly: LlamaLogic.Packages.dll

Represents an exception thrown by an IModel<TSelf> when it attempts to decode a resource that is not in the expected format

```
public sealed class UnexpectedFormatDecodingException : Exception, ISerializable
```

Inheritance

object ← Exception ← UnexpectedFormatDecodingException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.ReferenceEquals(object, object)

Constructors

UnexpectedFormatDecodingException()

Initializes a new UnexpectedFormatDecodingException

```
public UnexpectedFormatDecodingException()
```

UnexpectedFormatDecodingException(Exception?)

Initializes a new UnexpectedFormatDecodingException with the specified `innerException`

```
public UnexpectedFormatDecodingException(Exception? innerException)
```

Parameters

`innerException` Exception

UnexpectedFormatDecodingException(string)

Initializes a new UnexpectedFormatDecodingException with the specified `message`

```
public UnexpectedFormatDecodingException(string message)
```

Parameters

`message` string

UnexpectedFormatDecodingException(string, Exception?)

Initializes a new UnexpectedFormatDecodingException with the specified `message` and `innerException`

```
public UnexpectedFormatDecodingException(string message, Exception? innerException)
```

Parameters

`message` string

`innerException` Exception

Namespace LlamaLogic.Packages.Models.Data

Classes

DataModel

Represents a SimData or CombinedTuning resource (🔓)

DataModelObject

Represents an OBJECT reference in a SimData or CombinedTuning resource (🔓)

DataModelReference

Represents a reference in a SimData or CombinedTuning resource (🔓)

DataModelString

Represents a STRING8 or HASHEDSTRING8 reference in a SimData or CombinedTuning resource (🔓)

DataModelTable

Represents a table in a SimData or CombinedTuning resource (🔓)

DataModelVariant

Represents a VARIANT reference in a SimData or CombinedTuning resource (🔓)

DataModelVector

Represents a VECTOR reference in a SimData or CombinedTuning resource (🔓)

Enums

DataModelType

A type of data within a data resource

DataModelVersion

Known versions of the format for SimData and CombinedTuning resources

Class DataModel

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents a SimData or CombinedTuning resource (🔓)

```
public sealed class DataModel : Model, IModel<DataModel>, IModel
```

Inheritance

object ← Model ← DataModel

Implements

IModel<DataModel>, IModel

Inherited Members

Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Synchronous API Only

Due to performance considerations, this object model provides no asynchronous API. Thus, the caller will be blocked until the entire SimData or CombinedTuning resource is decoded by Decode(ReadOnlyMemory<byte>) or encoded by Encode(). Front-end developers are advised to wrap calls to either method with Run(Func<Task>) to avoid blocking the UI thread. The asynchronous API of DataBasePackedFile will do this automatically.

Eager Loading

When a SimData or CombinedTuning resource is passed to Decode(ReadOnlyMemory<byte>), the entire resource is decoded to compose a complete graph of all tables in the resource, including their metadata and records. For some CombinedTuning resources shipped by Maxis, this may result in a large number of tables and records being loaded into memory and take a significant amount of time.

Thread Safety

For performance reasons, this class and its supporting classes in this namespace are not thread-safe. If a caller has need for parallelized use of a DataModel, they are expected to manage synchronization on

their own.

Exposed Reference Structure

To grant callers maximum control, the structure of references from the SimData or CombinedTuning resource is preserved and unabstracted.

While the classes which implement DataModelReference have properties to produce useful compositions (e.g. the Value property of DataModelString will convert the null-terminated sequence of ASCII characters it references in a DataModelTable of CHAR8 values to a string), these conveniences are processed on-demand and only so long as the reference has remained valid. The original data being referenced remains stored by instances of DataModelTable as it appeared in the resource at the time it was decoded. All instances of DataModelReference monitor the row (or rows) of the DataModelTable they are referencing and will invalidate themselves if any constituent element is altered by the caller.

Encode() reacts to invalid references by writing the `RELOFFSET_NULL` value in their place. Callers can check to see if something they've done has invalidated a reference by checking its IsValid property.

Properties

this[int]

Gets the table at the specified `index`

```
public DataModelTable this[int index] { get; }
```

Parameters

`index` int

Property Value

DataModelTable

this[string]

Gets the table with the specified `name`

```
public DataModelTable this[string name] { get; }
```

Parameters

`name` string

Property Value

DataModelTable

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Tables

Gets the tables in this resource

```
public Collection<DataModelTable> Tables { get; }
```

Property Value

Collection<DataModelTable>

Version

Gets/sets the DataModelVersion of this resource

```
public DataModelVersion Version { get; set; }
```

Property Value

DataModelVersion

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔄 🏃)

```
public static DataModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

DataModel

Exceptions

UnexpectedFormatDecodingException

> The data appears to be a CombinedTuning resource in XML format

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

GetTable(string)

Gets the table with the specified `name`

```
public DataModelTable GetTable(string name)
```

Parameters

`name` string

Returns

DataModelTable

Class DataModelObject

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents an OBJECT reference in a SimData or CombinedTuning resource (🔓)

```
public class DataModelObject : DataModelReference
```

Inheritance

object ← DataModelReference ← DataModelObject

Derived

DataModelVariant

Inherited Members

DataModelReference.Column , DataModelReference.ColumnIndex , DataModelReference.IsValid , DataModelReference.Row , DataModelReference.RowIndex , DataModelReference.Table , DataModelReference.Type , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DataModelObject(DataModelTable, Index, Index, DataModelType)

Initializes a new DataModelObject (🔄 🏃)

```
public DataModelObject(DataModelTable table, Index rowIndex, Index columnIndex,
DataModelType type)
```

Parameters

`table` DataModelTable

The table containing the object being referenced

`rowIndex` Index

The row index within the containing table of the object being referenced

`columnIndex` Index

The column index within the containing table of the object being referenced

`type` DataModelType

The type of the object being referenced

Properties

NullReference

Gets the null reference DataModelObject

```
public static DataModelObject NullReference { get; }
```

Property Value

DataModelObject

Value

Gets/sets the value referenced by the object (💤)

```
public object? Value { get; set; }
```

Property Value

object

Class DataModelReference

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents a reference in a SimData or CombinedTuning resource (🔓)

```
public abstract class DataModelReference
```

Inheritance

object ← DataModelReference

Derived

DataModelObject, DataModelString, DataModelVector

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DataModelReference(DataModelTable, Index, Index, DataModelType)

Initializes a new DataModelReference (🔁 🏃)

```
protected DataModelReference(DataModelTable table, Index rowIndex, Index columnIndex,
DataModelType type)
```

Parameters

`table` DataModelTable

The table containing the structure being referenced

`rowIndex` Index

The row index within the containing table of the structure being referenced

`columnIndex` Index

The column index within the containing table of the structure being referenced

`type` DataModelType

The type of the structure being referenced

Properties

Column

Gets the column index within the table containing the reference of the reference

```
public int Column { get; }
```

Property Value

int

ColumnIndex

Gets the column index within the table containing the reference of the reference

```
public Index ColumnIndex { get; }
```

Property Value

Index

IsValid

Gets whether the reference has remained valid

```
public virtual bool IsValid { get; }
```

Property Value

bool

Row

Gets the row index within the table containing the reference of the reference (or -1 if the row has been removed)

```
public int Row { get; }
```

Property Value

int

RowIndex

Gets the row index within the table containing the reference of the reference (or -1 if the row has been removed)

```
public Index RowIndex { get; }
```

Property Value

Index

Table

Gets the table containing the structure being referenced

```
public DataModelTable Table { get; }
```

Property Value

DataModelTable

Type

Gets the type of the structure being referenced

```
public DataModelType Type { get; }
```

Property Value

DataModelType

Class DataModelString

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents a STRING8 or HASHEDSTRING8 reference in a SimData or CombinedTuning resource (🔓)

```
public sealed class DataModelString : DataModelReference
```

Inheritance

object ← DataModelReference ← DataModelString

Inherited Members

DataModelReference.Column , DataModelReference.ColumnIndex , DataModelReference.IsValid , DataModelReference.Row , DataModelReference.RowIndex , DataModelReference.Table , DataModelReference.Type , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DataModelString(DataModelTable, Index)

Initializes a new DataModelString (🔄 🏃)

```
public DataModelString(DataModelTable table, Index rowIndex)
```

Parameters

`table` DataModelTable

The table, a sub sequence of which, is the null-terminated string of CHAR8 values

`rowIndex` Index

The row index within the containing table of the string being referenced at which the string begins

Properties

Length

Gets the length of the string in ASCII characters, including the null terminator

```
public int Length { get; }
```

Property Value

int

NullReference

Gets the null reference DataModelString

```
public static DataModelString NullReference { get; }
```

Property Value

DataModelString

Range

Gets the range rows of the string

```
public Range Range { get; }
```

Property Value

Range

Value

Gets/sets the value of the string ( )

```
public string Value { get; set; }
```

Property Value

string

Methods

CreateForString(DataModel, string)

Creates a DataModelString based on a string

```
public static DataModelString CreateForString(DataModel dataModel, string str)
```

Parameters

`dataModel` DataModel

The model of which the string is a part

`str` string

The string

Returns

DataModelString

Class DataModelTable

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents a table in a SimData or CombinedTuning resource (🔓)

```
public sealed class DataModelTable
```

Inheritance

object ← DataModelTable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DataModelTable(string?, string?, uint?)

Represents a table in a SimData or CombinedTuning resource (🔓)

```
public DataModelTable(string? name, string? schemaName, uint? schemaHash)
```

Parameters

`name` string

`schemaName` string

`schemaHash` uint?

Properties

ColumnCount

Gets the number of columns

```
public int ColumnCount { get; }
```

Property Value

int

this[Index]

Gets/sets the values of the specified `rowIndex`

```
public object? this[Index rowIndex] { get; set; }
```

Parameters

`rowIndex` Index

Property Value

object

this[Index, Index]

Gets/sets the value at the specified `rowIndex` and `columnIndex`

```
public object? this[Index rowIndex, Index columnIndex] { get; set; }
```

Parameters

`rowIndex` Index

`columnIndex` Index

Property Value

object

this[Index, string]

Gets/sets the value of the specified `rowIndex` for column with the specified `columnName`

```
public object? this[Index rowIndex, string columnName] { get; set; }
```

Parameters

`rowIndex` Index

`columnName` string

Property Value

object

Name

Gets/sets the name of the table

```
public string? Name { get; set; }
```

Property Value

string

RawValues

Gets the raw values of the table

```
public IEnumerable<object?> RawValues { get; }
```

Property Value

IEnumerable<object>

RowCount

Gets the number of rows

```
public int RowCount { get; }
```

Property Value

int

SchemaHash

Gets/sets the unique identifier for the schema of the table

```
public uint? SchemaHash { get; set; }
```

Property Value

uint?

SchemaName

Gets/sets the name of the schema of the table

```
public string? SchemaName { get; set; }
```

Property Value

string

Methods

AddColumn(string?, DataModelType, ushort, IEnumerable<object?>?)

Adds a column with the specified `name`, `type`, `flags`, and `values`, and returns the index of the column

```
public int AddColumn(string? name, DataModelType type, ushort flags = 0, IEnumerable<object?
>? values = null)
```

Parameters

`name` string

`type` DataModelType

`flags` ushort

`values` IEnumerable<object>

Returns

int

AddRow(object?)

Adds the specified `row` and returns the index of the row

```
public int AddRow(object? row)
```

Parameters

`row` object

Returns

int

Get(Index)

Gets the values of the row at the specified `rowIndex`

```
public object? Get(Index rowIndex)
```

Parameters

`rowIndex` Index

Returns

object

Get(Index, Index)

Gets the value at the specified `rowIndex` and `columnIndex`

```
public object? Get(Index rowIndex, Index columnIndex)
```

Parameters

`rowIndex` Index

`columnIndex` Index

Returns

object

Get(Index, string)

Gets the value at the specified `rowIndex` for the column with the specified `columnName`

```
public object? Get(Index rowIndex, string columnName)
```

Parameters

`rowIndex` Index

`columnName` string

Returns

object

GetColumnFlags(Index)

Gets the flags of the column at the specified `index`

```
public ushort GetColumnFlags(Index index)
```

Parameters

`index` Index

Returns

ushort

GetColumnFlags(string)

Gets the flags of the column with the specified `columnName`

```
public ushort GetColumnFlags(string columnName)
```

Parameters

`columnName` string

Returns

ushort

GetColumnIndex(string)

Gets the index of the column with the specified `columnName`, or null if the column does not exist

```
public int? GetColumnIndex(string columnName)
```

Parameters

`columnName` string

Returns

int?

GetColumnName(Index)

Gets the name of the column at the specified `index`

```
public string? GetColumnName(Index index)
```

Parameters

`index` Index

Returns

string

GetColumnType(Index)

Gets the DataModelType of the column at the specified `index`

```
public DataModelType GetColumnType(Index index)
```

Parameters

`index` Index

Returns

DataModelType

GetColumnType(string)

Gets the DataModelType of the column with the specified `columnName`

```
public DataModelType GetColumnType(string columnName)
```

Parameters

`columnName` string

Returns

DataModelType

GetColumnValues(Index)

Gets all the values for the column at the specified `index`

```
public IReadOnlyList<object?> GetColumnValues(Index index)
```

Parameters

`index` Index

Returns

IReadOnlyList<object>

GetColumnValues(Index, Index, Predicate<object?>)

Gets the values for the specified `columnIndex` starting at the specified `startingRowIndex` for as long as the `takeWhile` predicate returns true

```
public IReadOnlyList<object?> GetColumnValues(Index columnIndex, Index startingRowIndex,
Predicate<object?> takeWhile)
```

Parameters

`columnIndex` Index

`startingRowIndex` Index

`takeWhile` Predicate<object>

Returns

IReadOnlyList<object>

GetColumnValues(Index, Range)

Gets the values for the column at the specified `columnIndex` for the specified `rows`

```
public IReadOnlyList<object?> GetColumnValues(Index columnIndex, Range rows)
```

Parameters

`columnIndex` Index

`rows` Range

Returns

IReadOnlyList<object>

GetColumnValues(string)

Gets all the values for the column with the specified `columnName`

```
public IReadOnlyList<object?> GetColumnValues(string columnName)
```

Parameters

`columnName` string

Returns

IReadOnlyList<object>

GetColumnValues(string, Index, Predicate<object?>)

Gets the values for the column with the specified `columnName` starting at the specified `startingRowIndex` for as long as the `takeWhile` predicate returns true

```
public IReadOnlyList<object?> GetColumnValues(string columnName, Index startingRowIndex,
Predicate<object?> takeWhile)
```

Parameters

`columnName` string

`startingRowIndex` Index

`takeWhile` Predicate <object>

Returns

IReadOnlyList <object>

GetColumnValues(string, Range)

Gets the values for the column with the specified `columnName` for the specified `rows`

```
public IReadOnlyList<object?> GetColumnValues(string columnName, Range rows)
```

Parameters

`columnName` string

`rows` Range

Returns

IReadOnlyList <object>

GetRawValue(Index)

Gets the value of a raw table for the specified `row`

```
public object? GetRawValue(Index row)
```

Parameters

`row` Index

Returns

object

GetRawValues(int, int)

Gets the values of a raw table for the specified number of `rows` beginning at the specified `rowIndex`

```
public IReadOnlyList<object?> GetRawValues(int rowIndex, int rows)
```

Parameters

`rowIndex` int

`rows` int

Returns

IReadOnlyList <object>

GetRawValues(Range)

Gets the values of a raw table for the specified `rows`

```
public IReadOnlyList<object?> GetRawValues(Range rows)
```

Parameters

`rows` Range

Returns

IReadOnlyList<object>

InsertColumn(Index, string?, DataModelType, ushort, IEnumerable<object?>?)

Inserts a column with the specified `name`, `type`, and `flags` at the specified `index`

```
public void InsertColumn(Index index, string? name, DataModelType type, ushort flags = 0,
IEnumerable<object?>? values = null)
```

Parameters

`index` Index

`name` string

`type` DataModelType

`flags` ushort

`values` IEnumerable<object>

InsertRow(Index, object?)

Inserts the specified `row` at the specified `index`

```
public void InsertRow(Index index, object? row)
```

Parameters

`index` Index

`row` object

RemoveColumn(Index)

Removes the column at the specified `index`

```
public void RemoveColumn(Index index)
```

Parameters

`index` Index

RemoveColumn(string)

Removes the column with the specified `columnName` and returns the index at which the column was removed

```
public int RemoveColumn(string columnName)
```

Parameters

`columnName` string

Returns

int

RemoveRow(Index)

Removes the row at the specified `index`

```
public void RemoveRow(Index index)
```

Parameters

`index` Index

Set(Index, Index, object?)

Sets the value at the specified `rowIndex` and `columnIndex`

```
public void Set(Index rowIndex, Index columnIndex, object? value)
```

Parameters

`rowIndex` Index

`columnIndex` Index

`value` object

Set(Index, object?)

Sets the `row` at the specified `index`

```
public void Set(Index index, object? row)
```

Parameters

`index` Index

`row` object

Set(Index, string, object?)

Sets the value of the specified `rowIndex` for column with the specified `columnName`

```
public void Set(Index rowIndex, string columnName, object? value)
```

Parameters

`rowIndex` Index

`columnName` string

`value` object

SetColumnFlags(Index, ushort)

Sets the flags of the column at the specified `index`

```
public void SetColumnFlags(Index index, ushort flags)
```

Parameters

`index` Index

`flags` ushort

SetColumnFlags(string, ushort)

Sets the flags of the column with the specified `columnName`

```
public void SetColumnFlags(string columnName, ushort flags)
```

Parameters

`columnName` string

`flags` ushort

SetColumnName(Index, string?)

Sets the name of the column at the specified `index`

```
public void SetColumnName(Index index, string? name)
```

Parameters

`index` Index

`name` string

SetRawValue(Index, object?)

Sets the `value` of a raw table for the specified `row`

```
public void SetRawValue(Index row, object? value)
```

Parameters

`row` Index

`value` object

SetRawValues(Range, IEnumerable<object?>)

Replaces the specified `rows` in a raw table with the specified `values`, returning the length of `values`

```
public int SetRawValues(Range rows, IEnumerable<object?> values)
```

Parameters

`rows` Range

`values` IEnumerable<object>

Returns

int

Enum DataModelType

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

A type of data within a data resource

```
public enum DataModelType
```

Fields

`BOOL = 0`

A bool expressed with an entire byte (`0` is false, any other value is true)

`CHAR8 = 1`

A single ASCII character

`FLOAT = 10`

A float

`FLOAT2 = 15`

You might think it's a double, but nope: two floats

`FLOAT3 = 16`

An obvious trend is developing, for this is three floats

`FLOAT4 = 17`

(You know when, every once in a while, there's a great trilogy and someone messes it up?) Here are your four floats

`HASHEDSTRING8 = 12`

An int offset to a null-terminated ASCII string and its accompanying Fnv32 hash

`INT16 = 4`

A short

`INT32 = 6`

An int

`INT64 = 8`

A long

`INT8 = 2`

An sbyte

`LOCKEY = 20`

A four-byte key used to uniquely identify localized strings in The Sims 4 (typically the Fnv32 hash of the string in its original language)

`OBJECT = 13`

An int offset to an object

`RESOURCEKEY = 19`

A ResourceKey laid out by FullInstance, then Type, then Group

`STRING8 = 11`

An int offset to a null-terminated ASCII string

`TABLESETREFERENCE = 18`

It's not just about good manners from "How to Serve and Not Be Served", it's really about this ulong

Seriously, though, why have these when we already have UINT64, you ask?

> you're right that table set refs are just uint64s, but it is important to distinguish them because they mean different things. int64s are just literal numbers, table set refs are tuning IDs. so, when the game loads these, it knows that uint64 is just a number, so leave it alone, whereas a table set ref is tuning, so it needs to look that tuning up and use it as the value there rather than the number itself

— frankk, August 16, 2024 in Lot 51's Discord

`UINT16 = 5`

A ushort

`UINT32 = 7`

A uint

`UINT64 = 9`

A ulong

`UINT8 = 3`

A byte

`UNDEFINED = 22`

?

`VARIANT = 21`

An int offset to an object accompanied by a uint type hash

`VECTOR = 14`

An int offset to an array of objects and a uint that is its size

Class DataModelVariant

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents a VARIANT reference in a SimData or CombinedTuning resource (🔓)

```
public sealed class DataModelVariant : DataModelObject
```

Inheritance

object ← DataModelReference ← DataModelObject ← DataModelVariant

Inherited Members

DataModelObject.NullReference , DataModelObject.Value , DataModelReference.Column , DataModelReference.ColumnIndex , DataModelReference.IsValid , DataModelReference.Row , DataModelReference.RowIndex , DataModelReference.Table , DataModelReference.Type , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DataModelVariant(DataModelTable, Index, Index, DataModelType, uint)

Initializes a new DataModelVariant (🔄 🏃)

```
public DataModelVariant(DataModelTable table, Index rowIndex, Index columnIndex,
DataModelType type, uint typeHash)
```

Parameters

`table` DataModelTable

The table containing the variant being referenced

`rowIndex` Index

The row index within the containing table of the variant being referenced

`columnIndex` Index

The column index within the containing table of the variant being referenced

`type` DataModelType

The type of the variant being referenced

`typeHash` uint

The type hash of the variant

Properties

TypeHash

Gets/sets the type hash of the variant

```
public uint TypeHash { get; set; }
```

Property Value

uint

Methods

CreateNullReference(uint)

Create a DataModelVariant null reference

```
public static DataModelVariant CreateNullReference(uint typeHash)
```

Parameters

`typeHash` uint

The type hash of the variant

Returns

DataModelVariant

Class DataModelVector

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Represents a VECTOR reference in a SimData or CombinedTuning resource (🔓)

```
public sealed class DataModelVector : DataModelReference
```

Inheritance

object ← DataModelReference ← DataModelVector

Inherited Members

DataModelReference.Column , DataModelReference.ColumnIndex , DataModelReference.Row , DataModelReference.RowIndex , DataModelReference.Table , DataModelReference.Type , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DataModelVector(DataModelTable, Index, DataModelType, int)

Initializes a new DataModelVector (🔄 🏃)

```
public DataModelVector(DataModelTable table, Index rowIndex, DataModelType type, int count)
```

Parameters

`table` DataModelTable

The table, a sub sequence of which, is the vector

`rowIndex` Index

The row index within the containing table of the vector being referenced at which the vector begins

`type` DataModelType

The type of the objects being referenced

`count` int

The number of elements in the vector

Properties

Count

Gets the number of elements in the vector

```
public int Count { get; }
```

Property Value

int

Indicies

Gets the indicies of the vector

```
public IEnumerable<int> Indicies { get; }
```

Property Value

IEnumerable <int>

IsValid

Gets whether the reference has remained valid

```
public override bool IsValid { get; }
```

Property Value

bool

Range

Gets the range rows of the vector

```
public Range Range { get; }
```

Property Value

Range

Values

Gets/sets the sequence of values referenced by the vector (💤)

```
public IEnumerable<object?> Values { get; set; }
```

Property Value

IEnumerable<object>

Methods

CreateForSequence(DataModel, DataModelType, IEnumerable<object?>)

Creates a DataModelVector based on a sequence of values

```
public static DataModelVector CreateForSequence(DataModel dataModel, DataModelType type,
IEnumerable<object?> sequence)
```

Parameters

`dataModel` DataModel

The model of which the vector is a part

`type` DataModelType

The type of elements in the vector

`sequence` IEnumerable<object>

The sequence of elements comprising the vector

Returns

DataModelVector

CreateNullReference(int)

Creates a DataModelVector null reference

```
public static DataModelVector CreateNullReference(int count = 0)
```

Parameters

`count` int

The number of elements (probably should be `0`)

Returns

DataModelVector

Enum DataModelVersion

Namespace: LlamaLogic.Packages.Models.Data

Assembly: LlamaLogic.Packages.dll

Known versions of the format for SimData and CombinedTuning resources

```
[SuppressMessage("Design", "CA1008: Enums should have zero value", Justification = "We do
not know what the proper name of such a value would be.")]
[SuppressMessage("Design", "CA1028: Enum Storage should be Int32", Justification = "This
is determined by the binary format created by Maxis. So sorry to clearly agitate you
so much.")]
public enum DataModelVersion : uint
```

Fields

`SmartSimOriginalEdition = 256`

The edition of the format which shipped with the original release of The Sims 4 on September 2, 2014

`VariantSupport = 257`

The edition of the format which shipped with The Sims 4 (v1.5.139) on February 17, 2015, introducing support for VARIANT and allowing the creation of CombinedTuning resources

Namespace LlamaLogic.Packages.Models.GlobalManifest

Classes

GlobalModsManifestModel

Represents the global mods manifest SnippetTuning resource (🔓)

GlobalModsManifestModelManifestedModFile

Represents a manifested mod in a global mods manifest SnippetTuning resource (🔓)

Class GlobalModsManifestModel

Namespace: LlamaLogic.Packages.Models.GlobalManifest

Assembly: LlamaLogic.Packages.dll

Represents the global mods manifest SnippetTuning resource (🔓)

```
public sealed class GlobalModsManifestModel : Model, IModel<GlobalModsManifestModel>,
IModel, IXmlSerializable
```

Inheritance

object ← Model ← GlobalModsManifestModel

Implements

IModel<GlobalModsManifestModel>, IModel, IXmlSerializable

Inherited Members

Model.GetName(Stream) , Model.GetNameAsync(Stream, CancellationToken) , Model.GetName(ResourceType, Stream) , Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

This model allows agents to easily create, read, and update the global mods manifest, which itself is an index of the player's installed packs and manifested mod files easily acessible to script mods.

Properties

DisabledPacks

Gets the collection of pack codes that the launcher has specified as disabled

```
public Collection<string> DisabledPacks { get; }
```

Property Value

Collection<string>

InstalledPacks

Gets the collection of pack codes for this game installation's installed packs

```
public Collection<string> InstalledPacks { get; }
```

Property Value

Collection<string>

ManifestedModFiles

Gets the collection of manifested mod files in the player's Mods folder

```
public Collection<GlobalModsManifestModelManifestedModFile> ManifestedModFiles { get; }
```

Property Value

Collection<GlobalModsManifestModelManifestedModFile>

ResourceKey

Gets the ResourceKey always used by the global mods manifest

```
public static ResourceKey ResourceKey { get; }
```

Property Value

ResourceKey

ResourceName

Gets the name of this resource if it has one

```
public override string? ResourceName { get; }
```

Property Value

string

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet<ResourceType>

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔄 🏃)

```
public static GlobalModsManifestModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

GlobalModsManifestModel

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

Class GlobalModsManifestModelManifestedModFile

Namespace: LlamaLogic.Packages.Models.GlobalManifest

Assembly: LlamaLogic.Packages.dll

Represents a manifested mod in a global mods manifest SnippetTuning resource (🔓)

```
public sealed class GlobalModsManifestModelManifestedModFile : IXmlSerializable
```

Inheritance

object ← GlobalModsManifestModelManifestedModFile

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Hashes

Gets the union of SubsumedHashes and the agent's calculated manifest hash for the mod file

```
public HashSet<ImmutableArray<byte>> Hashes { get; }
```

Property Value

HashSet<ImmutableArray<byte>>

ManifestKey

Gets/sets the ResourceKey of the mod manifest SnippetTuning of the mod file if it's a package

```
public ResourceKey? ManifestKey { get; set; }
```

Property Value

ResourceKey?

ManifestTuningName

Gets the tuning name of the mod manifest SnippetTuning of the mod file if it's a package

```
public string? ManifestTuningName { get; set; }
```

Property Value

string

ModsFolderPath

Gets/sets the relative path from the root of the Mods folder to the mod file

```
public string ModsFolderPath { get; set; }
```

Property Value

string

Namespace LlamaLogic.Packages.Models.ModFileManifest

Classes

ModFileManifestModel

Represents a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelRecommendedPack

Represents a recommended mod in a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelRepurposedLanguage

Represents a repurposed language in a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelRequiredMod

Represents a required mod in a mod file manifest SnippetTuning resource (🔓)

ModFileManifestModelTranslator

Represents the contribution of a translator in a mod file manifest SnippetTuning resource (🔓)

Class ModFileManifestModel

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModel : Model, IModel<ModFileManifestModel>,
IModel, IXmlSerializable
```

Inheritance

object ← Model ← ModFileManifestModel

Implements

IModel<ModFileManifestModel>, IModel, IXmlSerializable

Inherited Members

Model.GetName(ResourceType, Stream) ,
Model.GetNameAsync(ResourceType, Stream, CancellationToken) , object.Equals(object) ,
object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.ReferenceEquals(object, object)

Remarks

This model allows callers to easily create, read, and update a mod file manifest. These manifests are a format sponsored by the Llama Logic team to permit creators to specify the dependency requirements of their mods.

Use in Mod Package (`.package`) files

When originally creating packages, place one and only one mod file manifest SnippetTuning resource in the package and use sufficiently unique values for its FullInstance and TuningName. If an external application encounters multiple SnippetTuning manifest resources in a package, it must assume that a manifest un-aware tool has merged multiple manifested packages and process each manifest accordingly.

Avoid using Group `0x00000000` as this is informally reserved for Maxis. It would be a bizarre miracle if they started incorporating dependency information using a community format. Regardless, let's not make them feel unwelcome and hold their spot for them.

Use in Script Mod Archive (`.ts4script`) files

Because these files are just ZIP archives with a different extension and their contents are not flatly merged by the game as the contents of package files are, there is no risk of collision. Also, since they need not contend with game resource management, they need not be in an arcane format like XML, so they are in YAML instead. The static parsing methods of this type expect YAML representations of manifests and the ToString() method produces them. Name your mod file manifest resource `llamalogic.modfilemanifest.yml` and put it in the root of your `.ts4script` archive.

Properties

ContactEmail

Gets/sets the email address which can be used to contact this mod's creator

```
[YamlMember(Order = 7, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? ContactEmail { get; set; }
```

Property Value

string

ContactUrl

Gets/sets the URL which can be used to contact this mod's creator

```
[YamlMember(Order = 8, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? ContactUrl { get; set; }
```

Property Value

Uri

Creators

Gets the names of the creators of the mod

```
[YamlMember(Order = 3, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
```

```
public Collection<string> Creators { get; }
```

Property Value

Collection<string>

Description

Gets/sets the description of the mod

```
[YamlMember(Order = 2, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? Description { get; set; }
```

Property Value

string

ElectronicArtsPromoCode

Gets/sets the promo code it is suggested the player use during check out in the EA Store if purchasing a pack for use with this mod

```
[YamlMember(Order = 21, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? ElectronicArtsPromoCode { get; set; }
```

Property Value

string

ExcludedEntries

Gets the names of entries in a script mod archive to be excluded from the Hash

```
[YamlMember(Order = 15, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public HashSet<string> ExcludedEntries { get; }
```

Property Value

HashSet<string>

Exclusivities

Gets the globally unique names of the exclusivities of this mod, causing it to be incompatible with other mods which share one or more of them

```
[YamlMember(Order = 18, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Exclusivities { get; }
```

Property Value

Collection<string>

Features

Gets the names of the features unique to this mod which it offers to other mods as a dependency

```
[YamlMember(Order = 17, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Features { get; }
```

Property Value

Collection<string>

FundingUrl

Gets/sets the URL to which players can go to offer support to the mod creator

```
[YamlMember(Order = 6, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? FundingUrl { get; set; }
```

Property Value

Uri

Hash

Gets/sets the hash of the mod file

```
[YamlMember(Order = 13, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public ImmutableArray<byte> Hash { get; set; }
```

Property Value

ImmutableArray<byte>

HashResourceKeys

Gets the resource keys for the resources included in the Hash

```
[YamlMember(Order = 14, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public HashSet<ResourceKey> HashResourceKeys { get; }
```

Property Value

HashSet<ResourceKey>

IncompatiblePacks

Gets the list of pack codes identifying the packs incompatible with this mod (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 22, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> IncompatiblePacks { get; }
```

Property Value

Collection<string>

MessageToTranslators

Gets/sets the message to translators by this mod's creator

```
[YamlMember(Order = 10, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? MessageToTranslators { get; set; }
```

Property Value

string

Name

Gets/sets the name of the mod

```
[YamlMember(Order = 1, DefaultValuesHandling = DefaultValuesHandling.OmitDefaults)]
public required string Name { get; set; }
```

Property Value

string

RecommendedPacks

Gets the list of pack codes identifying the packs required by this mod (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 20, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelRecommendedPack> RecommendedPacks { get; }
```

Property Value

Collection <ModFileManifestModelRecommendedPack>

RepurposedLanguages

Gets the list of languages repurposed by this mod

```
[YamlMember(Order = 12, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelRepurposedLanguage> RepurposedLanguages { get; }
```

Property Value

Collection<ModFileManifestModelRepurposedLanguage>

RequiredMods

Gets the list of mods required by this mod

```
[YamlMember(Order = 23, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelRequiredMod> RequiredMods { get; }
```

Property Value

Collection<ModFileManifestModelRequiredMod>

RequiredPacks

Gets the list of pack codes identifying the packs required by this mod (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 19, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> RequiredPacks { get; }
```

Property Value

Collection<string>

ResourceName

Gets the name of this resource if it has one

```
[YamlIgnore]
public override string? ResourceName { get; }
```

Property Value

string

SubsumedHashes

Gets the hashes of previous versions of this mod in for which I stand even though my hash is different

```
[YamlMember(Order = 16, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public HashSet<ImmutableArray<byte>> SubsumedHashes { get; }
```

Property Value

HashSet <ImmutableArray <byte> >

SupportedTypes

Gets a list of resource types that this model supports

```
public static ISet<ResourceType> SupportedTypes { get; }
```

Property Value

ISet <ResourceType>

TranslationSubmissionUrl

Gets/sets the URL at which translators may submit translations to this mod's creator

```
[YamlMember(Order = 11, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? TranslationSubmissionUrl { get; set; }
```

Property Value

Uri

Translators

Gets the list of translators who have contributed translations to this mod

```
[YamlMember(Order = 9, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<ModFileManifestModelTranslator> Translators { get; }
```

Property Value

Collection<ModFileManifestModelTranslator>

TuningFullInstance

Gets the SnippetTuning decimal conversion of the FullInstance for the mod file manifest

```
[YamlMember(Order = -2, DefaultValuesHandling = DefaultValuesHandling.OmitDefaults)]
public ulong TuningFullInstance { get; set; }
```

Property Value

ulong

TuningName

Gets the SnippetTuning name for the mod file manifest

```
[YamlMember(Order = -1, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? TuningName { get; set; }
```

Property Value

string

Url

Gets/sets the URL to which players can go to find more information about this mod

```
[YamlMember(Order = 5, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? Url { get; set; }
```

Property Value

Uri

Version

Gets/sets the version of this mod

```
[YamlMember(Order = 4, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? Version { get; set; }
```

Property Value

string

Methods

Decode(ReadOnlyMemory<byte>)

Decodes the resource in raw format to produce an operable model (🔁 🏃)

```
public static ModFileManifestModel Decode(ReadOnlyMemory<byte> data)
```

Parameters

`data` ReadOnlyMemory<byte>

Returns

ModFileManifestModel

DeleteModFileManifests(ZipFile)

Deletes all mod file manifest resources in the specified `scriptMod`

```
public static void DeleteModFileManifests(ZipFile scriptMod)
```

Parameters

`scriptMod` ZipFile

DeleteModFileManifests(DataBasePackedFile)

Deletes all mod file manifest resources in the specified `package`

```
public static void DeleteModFileManifests(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

DeleteModFileManifestsAsync(DataBasePackedFile)

Deletes all mod file manifest resources in the specified `package`, asynchronously

```
public static Task DeleteModFileManifestsAsync(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

Task

Encode()

Encodes the resource model into raw format (🔄 🏃)

```
public override ReadOnlyMemory<byte> Encode()
```

Returns

ReadOnlyMemory<byte>

GetFileSha256Hash(string)

Gets the SHA 256 hash of the content of the file located at `filePath`

```
public static ImmutableArray<byte> GetFileSha256Hash(string filePath)
```

Parameters

`filePath` string

Returns

ImmutableArray<byte>

GetFileSha256HashAsync(string, CancellationToken)

Gets the SHA 256 hash of the content of the file located at `filePath` asynchronously

```
public static Task<ImmutableArray<byte>> GetFileSha256HashAsync(string filePath,
CancellationToken cancellationToken = default)
```

Parameters

`filePath` string

`cancellationToken` CancellationToken

Returns

Task<ImmutableArray<byte>>

GetModFileHash(ZipFile, HashSet<string>)

Gets the hash for the specified `scriptMod`, excluding entries with the names specified by `excludedEntries`

```
public static ImmutableArray<byte> GetModFileHash(ZipFile scriptMod,
HashSet<string> excludedEntries)
```

Parameters

`scriptMod` ZipFile

`excludedEntries` HashSet<string>

Returns

ImmutableArray<byte>

GetModFileHash(DataBasePackedFile, HashSet<ResourceKey>)

Gets the hash for the specified `package` using the specified `hashResourceKeys`

```
public static ImmutableArray<byte> GetModFileHash(DataBasePackedFile package,
HashSet<ResourceKey> hashResourceKeys)
```

Parameters

`package` DataBasePackedFile

`hashResourceKeys` HashSet<ResourceKey>

Returns

ImmutableArray<byte>

GetModFileHashAsync(DataBasePackedFile, HashSet<ResourceKey>, CancellationToken)

Gets the hash for the specified `package` using the specified `hashResourceKeys`, asynchronously

```
public static Task<ImmutableArray<byte>> GetModFileHashAsync(DataBasePackedFile package,
HashSet<ResourceKey> hashResourceKeys, CancellationToken cancellationToken = default)
```

Parameters

`package` DataBasePackedFile

`hashResourceKeys` HashSet<ResourceKey>

`cancellationToken` CancellationToken

Returns

Task<ImmutableArray<byte>>

GetModFileManifest(ZipFile)

Gets the mod file manifest for the specified `scriptMod`, if it has one

```
public static ModFileManifestModel? GetModFileManifest(ZipFile scriptMod)
```

Parameters

`scriptMod` ZipFile

Returns

ModFileManifestModel

GetModFileManifest(DataBasePackedFile)

Gets the mod file manifest for the specified `package`, if it has one

```
public static ModFileManifestModel? GetModFileManifest(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

ModFileManifestModel

GetModFileManifestAndKey(DataBasePackedFile)

Gets the mod file manifest for the specified `package` and its ResourceKey, if the `package` has one

```
public static (ResourceKey, ModFileManifestModel?)
GetModFileManifestAndKey(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

(ResourceKey, ModFileManifestModel)

GetModFileManifestAndKeyAsync(DataBasePackedFile)

Gets the mod file manifest for the specified `package`, asynchronously, if the `package` has one

```
public static Task<(ResourceKey, ModFileManifestModel?)>
GetModFileManifestAndKeyAsync(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

Task<(ResourceKey, ModFileManifestModel)>

GetModFileManifestAsync(ZipFile)

Gets the mod file manifest for the specified `scriptMod`, asynchronously, if it has one

```
public static Task<ModFileManifestModel?> GetModFileManifestAsync(ZipFile scriptMod)
```

Parameters

`scriptMod` ZipFile

Returns

Task<ModFileManifestModel>

GetModFileManifestAsync(DataBasePackedFile)

Gets the mod file manifest for the specified `package`, asynchronously, if it has one

```
public static Task<ModFileManifestModel?> GetModFileManifestAsync(DataBasePackedFile
package)
```

Parameters

`package` DataBasePackedFile

Returns

Task<ModFileManifestModel>

GetModFileManifests(DataBasePackedFile)

Gets the mod file manifests for the specified `package` (for use by editors checking for packages which may have been merged by manifest unaware tooling)

```
public static IReadOnlyDictionary<ResourceKey, ModFileManifestModel>
GetModFileManifests(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

IReadOnlyDictionary<ResourceKey, ModFileManifestModel>

GetModFileManifestsAsync(DataBasePackedFile)

Gets the mod file manifests for the specified `package`, asynchronously (for use by editors checking for packages which may have been merged by manifest unaware tooling)

```
public static Task<IReadOnlyDictionary<ResourceKey, ModFileManifestModel>>
GetModFileManifestsAsync(DataBasePackedFile package)
```

Parameters

`package` DataBasePackedFile

Returns

Task<IReadOnlyDictionary<ResourceKey, ModFileManifestModel>>

GetName(Stream)

Gets the name of a resource from its raw data, if it has one (🔄 💤)

```
public static string? GetName(Stream stream)
```

Parameters

`stream` Stream

Returns

string

GetNameAsync(Stream, CancellationToken)

Gets the name of a resource from its raw data, if it has one, asynchronously (🔄 💤)

```
public static Task<string?> GetNameAsync(Stream stream, CancellationToken cancellationToken
= default)
```

Parameters

`stream` Stream

`cancellationToken` CancellationToken

Returns

Task<string>

Parse(string)

Parses a string into a ModFileManifestModel

```
public static ModFileManifestModel Parse(string s)
```

Parameters

`s` string

The string to parse

Returns

ModFileManifestModel

Exceptions

FormatException

`s` is not in the correct format

Parse(string, IFormatProvider?)

Parses a string into a ModFileManifestModel

```
public static ModFileManifestModel Parse(string s, IFormatProvider? provider)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

Returns

ModFileManifestModel

Exceptions

FormatException

`s` is not in the correct format

SetModFileManifest(ZipFile, ModFileManifestModel)

Set the specified `manifest` for the specified `scriptMod`

```
public static void SetModFileManifest(ZipFile scriptMod, ModFileManifestModel manifest)
```

Parameters

`scriptMod` ZipFile

`manifest` ModFileManifestModel

SetModFileManifest(DataBasePackedFile, ModFileManifestModel)

Set the specified `manifest` for the specified `package`

```
public static void SetModFileManifest(DataBasePackedFile package,
ModFileManifestModel manifest)
```

Parameters

`package` DataBasePackedFile

`manifest` ModFileManifestModel

SetModFileManifestAsync(ZipFile, ModFileManifestModel)

Set the specified `manifest` for the specified `scriptMod`, asynchronously

```
public static Task SetModFileManifestAsync(ZipFile scriptMod, ModFileManifestModel manifest)
```

Parameters

`scriptMod` ZipFile

`manifest` ModFileManifestModel

Returns

Task

SetModFileManifestAsync(DataBasePackedFile, ModFileManifestModel)

Set the specified `manifest` for the specified `package`, asynchronously

```
public static Task SetModFileManifestAsync(DataBasePackedFile package,
ModFileManifestModel manifest)
```

Parameters

`package` DataBasePackedFile

`manifest` ModFileManifestModel

Returns

Task

ToString()

Generates the YAML representation of the ModFileManifestModel

```
public override string ToString()
```

Returns

string

TryParse(string?, out ModFileManifestModel)

Tries to parse a string into a ModFileManifestModel

```
public static bool TryParse(string? s, out ModFileManifestModel result)
```

Parameters

`s` string

The string to parse

`result` ModFileManifestModel

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

TryParse(string?, IFormatProvider?, out ModFileManifestModel)

Tries to parse a string into a ResourceKey

```
public static bool TryParse(string? s, IFormatProvider? provider, out
ModFileManifestModel result)
```

Parameters

`s` string

The string to parse

`provider` IFormatProvider

An object that provides culture-specific formatting information about `s`

`result` ModFileManifestModel

When this method returns, contains the result of successfully parsing `s` or an undefined value on failure

Returns

bool

Class ModFileManifestModelRecommendedPack

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a recommended mod in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelRecommendedPack : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelRecommendedPack

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

PackCode

Gets/sets the pack code indicating the pack that is recommended (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 1)]
public string PackCode { get; set; }
```

Property Value

string

Reason

Gets/sets the reason for the pack recommendation

```
[YamlMember(Order = 2)]
public string Reason { get; set; }
```

Property Value

string

Class ModFileManifestModelRepurposedLanguage

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a repurposed language in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelRepurposedLanguage : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelRepurposedLanguage

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ActualLocale

Gets/sets the IETF BCP 47 language tag identifying the non-game-supported language this mod actually contains in the domain ordinarily reserved to GameLocale

```
[YamlMember(Order = 2)]
public string ActualLocale { get; set; }
```

Property Value

string

GameLocale

Gets/sets the IETF BCP 47 language tag identifying the game-supported language this mod is repurposing

```
[YamlMember(Order = 1)]
public string GameLocale { get; set; }
```

Property Value

string

Remarks

The IETF BCP 47 language tags currently supported by The Sims 4 at time of writing are:

Tag	Description
en-US	American English | English (United States) | Full instance prefix: `0x00`
zh-CN	简体中文 | Chinese (Simplified, Mainland China) | Full instance prefix: `0x01`
zh-TW	繁體中文 | Chinese (Traditional, Taiwan) | Full instance prefix: `0x02`
cz	Čeština | Czech | Full instance prefix: `0x03`
da	Dansk | Danish | Full instance prefix: `0x04`
nl	Nederlands | Dutch | Full instance prefix: `0x05`
fi	Suomi | Finnish | Full instance prefix: `0x06`
fr-FR	Français | French (France) | Full instance prefix: `0x07`
de	Deutsch | German | Full instance prefix: `0x08`
it	Italiano | Italian | Full instance prefix: `0x0b`
ja	日本語 | Japanese | Full instance prefix: `0x0c`
ko	한국어 | Korean | Full instance prefix: `0x0d`
nb	Norsk | Norwegian | Full instance prefix: `0x0e`
pl	Polski | Polish | Full instance prefix: `0x0f`

Tag	Description
pt-BR	Português | Portuguese (Brazil) | Full instance prefix: `0x11`
ru	Русский | Russian | Full instance prefix: `0x12`
es-ES	Español | Spanish (Spain) | Full instance prefix: `0x13`
sv	Svenska | Swedish | Full instance prefix: `0x15`

Class ModFileManifestModelRequiredMod

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents a required mod in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelRequiredMod : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelRequiredMod

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Creators

Gets the names of the creators of the dependency mod

```
[YamlMember(Order = 2, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> Creators { get; }
```

Property Value

Collection<string>

Hashes

Gets the hashes of the mod files that must be present in order for this dependency to be fulfilled

```
[YamlMember(Order = 7)]
public HashSet<ImmutableArray<byte>> Hashes { get; }
```

Property Value

HashSet<ImmutableArray<byte>>

IgnoreIfHashAvailable

Instructs agents to ignore fulfilling this dependency if this hash is present in the player's catalog of mods

```
[YamlMember(Order = 8, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public ImmutableArray<byte> IgnoreIfHashAvailable { get; set; }
```

Property Value

ImmutableArray<byte>

IgnoreIfHashUnavailable

Instructs agents to ignore fulfilling this dependency if this hash is not present in the player's catalog of mods

```
[YamlMember(Order = 9, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public ImmutableArray<byte> IgnoreIfHashUnavailable { get; set; }
```

Property Value

ImmutableArray<byte>

IgnoreIfPackAvailable

Instructs agents to ignore fulfilling this dependency if the pack identified by this pack code is present (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 10, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? IgnoreIfPackAvailable { get; set; }
```

Property Value

string

IgnoreIfPackUnavailable

Instructs agents to ignore fulfilling this dependency if the pack identified by this pack code is not present (e.g. "EP01" for Get to Work)

```
[YamlMember(Order = 11, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? IgnoreIfPackUnavailable { get; set; }
```

Property Value

string

Name

Gets/sets the name of the dependency mod

```
[YamlMember(Order = 1)]
public string Name { get; set; }
```

Property Value

string

RequiredFeatures

Gets the names of the features of the dependency mod which the dependent mod requires

```
[YamlMember(Order = 5, DefaultValuesHandling = DefaultValuesHandling.OmitEmptyCollections)]
public Collection<string> RequiredFeatures { get; }
```

Property Value

Collection<string>

RequirementIdentifier

Gets/sets an identifier that, when shared with a group of other dependency mods for a single dependent, indicates that only one member of the group need be present for the dependent to be satisfied

```
[YamlMember(Order = 6, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? RequirementIdentifier { get; set; }
```

Property Value

string

Url

Gets/sets the URL to which players can go to find more information about this dependency mod

```
[YamlMember(Order = 4, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public Uri? Url { get; set; }
```

Property Value

Uri

Version

Gets/sets the version of this dependency mod

```
[YamlMember(Order = 3, DefaultValuesHandling = DefaultValuesHandling.OmitNull)]
public string? Version { get; set; }
```

Property Value

string

Class ModFileManifestModelTranslator

Namespace: LlamaLogic.Packages.Models.ModFileManifest

Assembly: LlamaLogic.Packages.dll

Represents the contribution of a translator in a mod file manifest SnippetTuning resource (🔓)

```
public sealed class ModFileManifestModelTranslator : IXmlSerializable
```

Inheritance

object ← ModFileManifestModelTranslator

Implements

IXmlSerializable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Culture

Gets/sets the value of the Language property using a CultureInfo object

```
[YamlIgnore]
public CultureInfo? Culture { get; set; }
```

Property Value

CultureInfo

Language

Gets/sets the IETF BCP 47 language tag identifying the language to which this translator contributed

```
[YamlMember(Order = 2)]
public string Language { get; set; }
```

Property Value

string

Name

Gets/sets the name of the translator

```
[YamlMember(Order = 1)]
public string Name { get; set; }
```

Property Value

string